Household Expectations in Uncertain Times*

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Abstract

How does uncertainty influence expectation formation? Using the U.S. Survey of Consumer Expectations, we show that uncertainty has opposite effects on belief updating depending on its origin: higher prior uncertainty increases updating, while noisier new information reduces it. Households are qualitatively Bayesian, but systematically overreact relative to the rational benchmark. We estimate a belief-updating model that disciplines these deviations and delivers a tractable framework for separating the two sources of uncertainty from survey data. As high-frequency surveys become increasingly common across countries, our framework provides a real-time diagnostic of the type of uncertainty policymakers face.

Keywords: Expectations, inflation, household surveys, consumer beliefs, information frictions, uncertainty.

JEL Classification: D81, D83, D84, E31.

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Uncertainty has moved to the center of macroeconomic debate. Its consequences are well documented: higher uncertainty reduces consumption and investment, increases precautionary saving, and makes policy less effective (Bloom et al., 2018; Fermand et al., 2024; Coibion et al., 2024; Georgarakos et al., 2024). These effects operate in part through expectations, as uncertainty reduces the precision consumers attach to their forecasts. Yet we know relatively little about how uncertainty shapes the way agents process new information — even though the responsiveness of beliefs to news is central to macroeconomic dynamics, the transmission of monetary policy, and asset prices (Maćkowiak and Wiederholt, 2015; Angeletos and La'o, 2020; Kozlowski et al., 2020; Afrouzi and Yang, 2021).

Our key insight is that uncertainty is not a monolith. A household's uncertainty about future inflation is driven by two factors: first, greater volatility in economic fundamentals makes past information and thus the household's priors less reliable; second, new data signals may be noisier or harder to interpret, reducing their informativeness. These two sources are observationally similar: both raise subjective uncertainty in surveys, and standard measures conflate them. But their implications for belief dynamics are sharply different. When priors are more uncertain, households are more responsive to new information. When new information is noisy, they discount it and hold on to their prior beliefs. Understanding which source drives observed uncertainty is therefore essential for interpreting belief dynamics — yet existing empirical approaches typically treat uncertainty as a single object.

This paper develop a framework to separate these two sources empirically and estimate their effects on household belief formation. We use the U.S. Survey of Consumer Expectations (SCE), a monthly rotating panel that elicits both point forecasts and density forecasts of inflation from a large sample of households. The density forecasts provide individual-level measures of subjective uncertainty, which we decompose into prior uncertainty and new information noise using the structure of a general linear updating model. We estimate belief stickiness—the weight households place on prior beliefs relative to new information—from the cross-sectional correlation between prior and posterior forecasts. Our identification exploits within-group variation, which removes common components of the signal and isolates individual responsiveness to news.

We establish three sets of results. The first concerns the relationship between the sources of uncertainty and belief dynamics. We construct proxies for these two sources from individual posterior uncertainty and confirm that their effects align qualitatively with the Bayesian framework: a one-standard-deviation increase in prior uncertainty is associated with approximately 20 percent lower belief stickiness, while a comparable increase in new information noise is asscoiated with an increase of about 30 percent.

The second set of results concerns the rationality of the updating process. We compare the estimated weight households place on new information with the Bayesian benchmark implied by their subjective uncertainty. While households update in the direction predicted by Bayes' rule, they overweight new information relative to its noise: on average, they assign roughly 25 percent more weight to new information than the rational benchmark implies. This overreaction is consistent with models of diagnostic expectations (Bordalo et al., 2020) and overconfidence (Adam et al., 2025). Importantly, the bias varies systematically across the population. Younger and lower-numeracy households display the strongest overreaction, while older and higher-numeracy individuals update closer to the Bayesian benchmark. This heterogeneity suggests that differences in belief stickiness across groups reflect behavioral biases rather than differences in information quality.

The third contribution is methodological. Because the two sources of uncertainty move updating in opposite directions, observed changes in belief stickiness can serve as a diagnostic tool for identifying which source is driving observed changes in uncertainty. We formalize this by combining the estimated belief-formation model with observable survey moments—the updating gain and posterior uncertainty—to back out the latent components of uncertainty as functions of the data alone. The method requires only repeated cross-sections of density forecasts and prior beliefs, making it applicable wherever such data are available.

We apply this decomposition to the post-2020 period in the United States, which provides a compelling setting because it features a large, persistent increase in subjective uncertainty.[1] Our framework reveals two sharply distinct regimes. The rise in uncertainty at the onset of COVID-19 reflects higher prior uncertainty, consistent with a sudden increase in the macroeconomic volatility that rendered existing information obsolete. Accordingly, belief stickiness dropped from roughly 0.65 to 0.30 as households became far more responsive to new information. During the subsequent high-inflation period beginning in 2021, uncertainty remained elevated but for a different reason: new information became noisier, consistent with the growing difficulty of interpreting conflicting economic signals. Belief stickiness returned to its pre-pandemic level, consistent with households discounting the noisier news they received.

These two regimes carry distinct policy implications. When uncertainty stems from noisier information—because information costs have risen or the supply of reliable signals has deteriorated—clearer communication by central banks and policymakers can directly reduce it. When uncertainty reflects higher fundamental volatility that has rendered prior beliefs obsolete, communication strategies have limited traction, and broader stabilization measures are required. As high-frequency household expectation surveys become increasingly common across countries, the framework we develop provides a real-time diagnostic of the type of uncertainty policymakers face.

Contributions to the literature Our work connects to several strands of the literature. We contribute to the literature on the measurement and consequences of macroeconomic uncertainty (Bloom, 2009; Jurado et al., 2015; Baley and Blanco, 2019; Baker et al., 2024; Coibion et al., 2024; Georgarakos et al., 2024).[2]. A well-known limitation of existing measures is that they conflate distinct sources—such as macroeconomic volatility and information noise—that are conceptually different and have different implications (Kozeniauskas et al.,

[1]After the COVID-19 pandemic, the Survey of Consumer Expectations (SCE) recorded the largest and persistent rise in subjective uncertainty since the survey began over a decade prior (Armantier et al., 2021).

[2]For work measuring uncertainty with survey data, see also Manski (2018); Binder et al. (2022); Kumar et al. (2023); Fermand et al. (2024); Wang (2024); Binder et al. (2025)

2018). Our approach provides a tractable way to separate them using survey data alone. Gambetti et al. (2023) pursue a related goal using forecast disagreement. Our approach complements theirs by exploiting a moment — belief stickiness — whose response to each source of uncertainty is theoretically unambiguous, allowing for a sharper decomposition.

We also contribute to the empirical study of information frictions in expectation surveys (Mankiw and Reis, 2002; Coibion and Gorodnichenko, 2015a; Kohlhas and Walther, 2021; Benhima and Bolliger, 2022; Baley and Turén, 2023). Unlike most of this work, which focuses on professional forecasters, we study household-level data and exploit individual density forecasts to separately identify the two components of uncertainty. Our cross-sectional identification approach builds on Goldstein (2023), who documents a decline in belief stickiness among professional forecasters during early COVID-19. Pfäuti (2023) also measures time variation in belief stickiness using the Michigan Survey of Consumers, relating it to the level of inflation. We study a different mechanism: we show that belief stickiness depends on the composition of uncertainty, and that this relationship holds even after controlling for inflation levels.[3]

Third, we add to the literature testing deviations from rational expectations (Bordalo et al., 2020; Broer and Kohlhas, 2024; Adam et al., 2025). Existing tests typically require tracking individual forecasters over long horizons to detect predictability in forecast errors, limiting their application to professional forecaster panels. Our approach exploits cross-sectional variation, comparing the level of belief stickiness to the Bayesian benchmark implied by observed uncertainty. This enables the study of rationality deviations in consumer surveys with richer demographic heterogeneity.[4]

Finally, our findings complement the experimental literature on information treatments

[3] De Bruin et al. (2011) also examines subjective uncertainty in the SCE and documents that more uncertain consumers revise their beliefs more. We consider both posterior and prior uncertainty and estimate their impact on information rigidities. Baker et al. (2020) proxies uncertainty shocks with natural disasters and finds that they decrease both belief stickiness and forecast errors. We directly measure individual subjective uncertainty and show that the relation with belief stickiness depends on the source of uncertainty.

[4] Our approach also avoids a limitation of tests based on professional forecasters, which may conflate cognitive biases with strategic diversification or reputational incentives (?).

and expectations (Armantier et al., 2016; Cavallo et al., 2017; Armona et al., 2019; Roth and Wohlfart, 2020; Coibion et al., 2022; Link et al., 2023). A common finding is that treatments are more effective when prior beliefs are uncertain and signals more informative.[5] Rather than relying on experimental variation, we leverage *naturally occurring* variation in prior uncertainty and belief updating, which allows us to examine time variation in belief stickiness and address the external validity concerns inherent in RCTs. Our approach is complementary to Weber et al. (2025), who show that information treatments become less effective during periods of high inflation, potentially due to increased pre-treatment attention that lowers prior uncertainty (Mackowiak and Wiederholt, 2024). Our estimates confirm this mechanism outside the experimental setting, showing that prior precision is a robust predictor of belief stickiness in observational data.[6]

Roadmap The rest of the paper is organized as follows. Section 1 presents the general framework that guides our empirical strategy, which we implement in Section 2. Section 3 studies the mechanisms underlying the relation between belief updating and uncertainty. Section 4 applies our method to decompose the post-2020 uncertainty surge. Section 5 concludes.

1 Framework and data

This section introduces a general belief-updating framework that links (i) how strongly households revise their inflation expectations (the updating gain) and (ii) how uncertain they are about future inflation (subjective uncertainty). We then describe the Survey of Consumer

[5]This pattern is well-documented for inflation expectations. For other variables, however, the evidence is mixed: Fuster et al. (2022) find the opposite effect for housing price expectations, while Armona et al. (2019) and Conlon et al. (2018) find no significant effects of uncertainty in the housing and labor markets.

[6]A related literature studies endogenous information acquisition and attention allocation by consumers and firms (Roth et al., 2022; Mikosch et al., 2024; Link et al., 2024). Rather than analyzing the determinants of attention choices, we measure the outcome of that process—the quantity of information held—through subjective uncertainty. As we show in Section 3.1, in the Bayesian framework, belief stickiness depends solely on this quantity, regardless of its source.

Expectations (SCE) data and explain how we estimate time- and group-varying belief stickiness from forecast revisions.

1.1 A general belief updating framework

We present a general theoretical framework embedding different models of belief updating, which will guide our empirical strategy. Households perceive inflation x_t to follow an AR(1) process,

$$x_t = (1 - \rho)\mu_x + \rho x_{t-1} + u_t, \tag{1}$$

where $u_t \sim N(0, \sigma_u^2)$ captures fundamental innovations and μ_x is the long-run mean.

Signals. At time t, household i in group j forms beliefs about inflation h periods ahead, x_{t+h}, after observing a (possibly composite) private signal,[7]

$$s_{i,j,t} = x_{t+h} + e_{i,j,t}, \tag{2}$$

where the signal noise decomposes as $e_{i,j,t} = \eta_{i,j,t} + \omega_t$.

The term $\eta_{i,j,t} \sim N(0, \sigma^2_{\eta,j,t})$ is idiosyncratic (i.i.d. across households and time) and satisfies $\int^i \eta_{i,j,t}\, di = 0$ within a group-month; $\omega_t \sim N(0, \sigma^2_{\omega,t})$ is common across households at time t (i.i.d. across time). Let

$$\sigma^2_{e,j,t} \equiv \sigma^2_{\eta,j,t} + \sigma^2_{\omega,t} \tag{3}$$

denote the total perceived noise in new information for group j at time t.[8]

[7] Our assumption that the signal is unbiased and centered on the realization is without loss of generality under mild conditions. If agents instead observe $\tilde{s}_{i,j,t} = \alpha_1 + \alpha_2 x_{t+h} + \phi_{i,j,t}$ with $\phi_{i,j,t} \sim N(0, \sigma^2_{\phi,j,t})$, they can rescale to $s_{i,j,t} \equiv (\tilde{s}_{i,j,t} - \alpha_1)/\alpha_2 = x_{t+h} + e_{i,j,t}$ with $e_{i,j,t} \equiv \alpha_2^{-1}\phi_{i,j,t}$.

[8] Normality is standard in this literature and is shared by several belief-formation models used to interpret survey expectations (e.g. Coibion and Gorodnichenko, 2015b; Bordalo et al., 2020).

A linear updating rule and attention. Households revise beliefs using a general *linear* updating rule:

$$
\begin{aligned}
E_{i,j,t}[x_{t+h}] &= (1 - G_{j,t})\, E_{i,j,t-1}[x_{t+h}] + G_{j,t}\, s_{i,j,t}, \\
E_{i,j,t-1}[x_{t+h}] &= (1-\rho)\mu_x + \rho\, E_{i,j,t-1}[x_{t+h-1}],
\end{aligned}
\tag{4}
$$

where $G_{j,t} \in \mathbb{R}$ is the weight placed on new information (the *gain*), which we also refer to as *attention* (greater responsiveness to news). We call $1 - G_{j,t}$ *belief stickiness* (or information rigidity). The second line describes how last month's forecast at horizon $h-1$ maps into this month's prior for horizon h under the perceived AR(1).[9]

We leave unrestricted what determines $G_{j,t}$. The only maintained assumption is that posterior beliefs are a linear combination of a prior and a signal. This nests Bayesian rational updating (where $G_{j,t}$ equals the Kalman gain) but also accommodates behavioral departures in which the gain is distorted, time-varying, or group-dependent (see Appendix A).

Subjective uncertainty. Let $\sigma^2_{i,j,t+h,t}$ denote household i's *posterior* variance for x_{t+h} at time t, and $\sigma^2_{i,j,t+h,t-1}$ the corresponding *prior* variance (i.e. before observing the time-t signal). Under the updating rule (4), the second moment evolves as

$$
\sigma^2_{i,j,t+h,t} = (1 - G_{j,t})^2 \sigma^2_{i,j,t+h,t-1} + G^2_{j,t} \sigma^2_{e,j,t}. \tag{5}
$$

Equation (5) makes transparent that posterior uncertainty rises with both (i) worse prior information (higher $\sigma^2_{i,j,t+h,t-1}$) and (ii) noisier new information (higher $\sigma^2_{e,j,t}$).[10]

[9]The perceived AR(1) in the second line of (4) is used only as a horizon-shifting device to propagate priors across survey months (i.e., to map the respondent's lagged $(h-1)$-ahead forecast into a prior for the h-ahead forecast). We do not impose a richer latent-state law of motion for beliefs beyond this formulation. Moreover, we estimate the gain separately by forecast horizon; to keep notation light we suppress the horizon index (baseline: $h = 3$ years; robustness: $h = 1$ year in Appendix N).

[10]Equation (5) implies that, as long as information quality and belief stickiness is the same across households in group j, then belief uncertainty is also the same: $\sigma_{i,j,t+h,t} = \sigma_{j,t+h,t}$. Since in this framework, as in a large set of belief formation models, the source of variation in uncertainty is the same as in belief stickiness, it is not possible to estimate G_t by regressing posterior and prior uncertainty with simple OLS.

Terminology. We use *belief stickiness* to indicate $1 - G_{j,t}$ (Coibion and Gorodnichenko, 2012, 2015a). We refer to $G_{j,t}$ as *updating gain* (greater responsiveness to news). Another widely used term in the recent literature is "attention", but this may be confused with the endogenous-information literature, where "attention" typically denotes a *choice* that reduces signal noise $\sigma^2_{e,j,t}$ (Sims, 2003; Mackowiak and Wiederholt, 2009). Our empirical strategy remains agnostic about whether rigidity reflects optimal information choices or behavioral distortions. Finally, we do not interpret our 3-year expectations as long-run anchoring to a central bank target, which typically requires much longer horizons (e.g. Gürkaynak et al., 2005).

1.2 Data

Realized inflation. We measure realized inflation using the monthly CPI-U series (CPIAUCSL) from FRED. To match the survey wording, we compute year-over-year inflation.

Survey expectations. We use the Federal Reserve Bank of New York's *Survey of Consumer Expectations* (SCE), a monthly rotating panel of roughly 1,200 household heads that begins in late 2012 (Armantier et al., 2017). Respondents remain in the panel for up to 16 months. We use the core sample from 2013 onward through 2023, which contains monthly point forecasts and density forecasts of inflation, as well as demographics and background information. We keep point forecasts only when a meaningful density forecast is available and the point forecast lies within the support of the density forecast. We also trim extreme point forecasts within each month-horizon cell to limit outliers.

The SCE elicits (i) expected inflation over the next 12 months ("1-year") and (ii) expected inflation over the 12 months starting 24 months in the future ("3-years"). Our baseline horizon is 3 years; we use the 1-year horizon for robustness (Appendix N).

Expectation mean. We measure the posterior mean $E_{i,t}[x_{t+h}]$ using the respondent's point forecast at time t for horizon h. To construct the prior mean $E_{i,t-1}[x_{t+h}]$, we use the AR(1) mapping in the second line of (4), based on the respondent's lagged forecast at horizon $h-1$. We calibrate (ρ, μ_x) using an AR(1) estimated on inflation data (Appendix D and Table A.1). Our benchmark specification uses the estimated parameters $\hat{\rho} = 0.946$ and $\hat{\mu}_x = 2.3$ to construct prior expectations. In Appendix P we replicate our main results using alternative calibrations, with virtually the same results.

Expectation uncertainty. We measure posterior uncertainty using the standard deviation of the subjective density forecast. The SCE elicits probabilities across 10 bins spanning roughly $[-12\%, 12\%]$ (Appendix B). The FRBNY fits parametric distributions to each respondent's histogram and reports individual variances using the method of Engelberg et al. (2009) (see Armantier et al., 2017). We denote the posterior standard deviation by $\sigma_{i,t+h,t}$.[11] Using density forecasts has two advantages over dispersion-based proxies: it provides individual-level uncertainty and maps directly to the second-moment restriction (5).[12]

Sociodemographics The SCE reports gender, age, and race. We construct indicators for college completion and for income categories (below \$50k, \$50k–\$100k, above \$100k). The SCE also reports respondents' numeracy, based on their ability to answer questions about probabilities and compound interest (Lusardi, 2008). Finally, *tenure* measures the number of months the respondent has participated in the panel (1–16). Tenure may matter through "learning-through-survey" (Kim and Binder, 2023), so we include it in our heterogeneity analysis.

[11] As described above, we use point forecasts to capture the first moment of subjective beliefs and density forecasts to measure the second moment. This separation avoids mechanical correlations between the two, such as those arising from measurement error. Nonetheless, in Appendix O, we show that our main results are robust to using the mean of the density forecast as a measure of expected inflation.

[12] A potential concern is that the upper limit of the survey bins may be too low during periods of high inflation. In such cases, clustering of responses in the top bin could lead to measurement error in the uncertainty measure. However, in Appendix C we show that our bin-based uncertainty measure closely tracks an alternative based instead point forecasts rounding, as in Binder (2017).

2 Identifying belief stickiness from the cross-section

We estimate belief stickiness by comparing posterior and prior forecasts across households, following the cross-sectional approach in Goldstein (2023) and **?**. This approach is well-suited to the SCE compared to other approaches (e.g., Coibion and Gorodnichenko, 2015a) because it (i) does not require a long time series and (ii) remains valid in the presence of a common signal component ω_t in (2), which biases time-series identification strategies based on forecast errors.

Identification intuition. From (4) and (2), demeaning within each group-month removes the common components of the signal (the realization x_{t+h} and the common noise ω_t). This yields a cross-sectional relationship between revisions and priors that identifies stickiness.

Formally, letting $\bar{E}_{j,t}[x_{t+h}] \equiv \int^{i \in j} E_{i,j,t}[x_{t+h}]\, di$, we obtain

$$E_{i,j,t}[x_{t+h}] - \bar{E}_{j,t}[x_{t+h}] = (1 - G_{j,t})\Big(E_{i,j,t-1}[x_{t+h}] - \bar{E}_{j,t-1}[x_{t+h}]\Big) + G_{j,t}\, \eta_{i,j,t}. \tag{6}$$

Group-month estimation. Equation (6) implies that, within a group j and month t, the OLS slope in the regression

$$E_{i,j,t}[x_{t+h}] = \alpha_{j,t} + \beta_{j,t}\, E_{i,j,t-1}[x_{t+h}] + err_{i,j,t} \tag{7}$$

identifies belief stickiness:

$$\beta_{j,t} = 1 - G_{j,t}. \tag{8}$$

Intuitively, higher belief stickiness implies a higher correlation between posterior beliefs and prior beliefs (higher $\beta_{j,t}$), while lower belief stickiness implies a lower correlation between posterior beliefs and prior beliefs (lower $\beta_{j,t}$). Our stickiness measure $\beta_{j,t}$ captures the average responsiveness of expectations to new information, combining both the extensive margin (whether households update) and the intensive margin (how much they update conditional on updating).

2.1 Heterogeneity across households

Estimating equation (7) requires selecting the socioeconomic characteristics used to group households by similarity in information quality and belief stickiness. In this Section we abstract from time variation in belief updating and instead investigate how belief stickiness, $1 - G_j$, varies across observable household characteristics.

We estimate a pooled specification that allows the slope on prior expectations to vary with socioeconomic characteristics:

$$E_t^i[x_{t+h}] = \alpha + \beta_1 E_{t-1}^i[x_{t+h}] + \mathbf{X}_{i,t}\mathbf{B}_2 + E_{t-1}^i[x_{t+h}] \times \mathbf{X}_{i,t}\mathbf{B}_3 + \gamma_t + \beta_4(\gamma_t \times \mathbf{X}_{i,t}) + \epsilon_t^i, \quad (9)$$

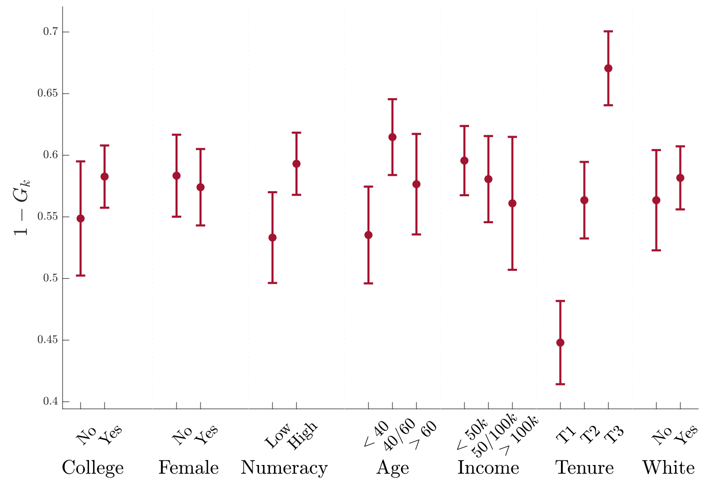
where $\mathbf{X}_{i,t}$ includes binary indicators for tenure terciles, education, age, income, numeracy, gender, and race. The interaction coefficients in $\mathbf{B}_3$ capture how stickiness varies with each characteristic.

We find that belief updating differs significantly across socioeconomic groups. Figure 1 plots belief stickiness by characteristic k (see detailed regression results in Table A.3). Households with longer tenure, a college degree, and higher numeracy exhibit significantly greater belief stickiness. In contrast, young respondents display lower belief stickiness.[13]

In the Bayesian framework, belief stickiness reflects the perceived precision of incoming information. In this framework, our findings would imply that younger, less experienced, or less skilled individuals receive more accurate signals. But households may depart from Bayesian updating, and the observed heterogeneity may reflect behavioral biases rather than differences in information quality. Sections 3–4 investigate this by using direct measures of perceived information accuracy.

[13]For a detailed analysis of attention heterogeneity in consumer and professional forecasters surveys, see Boccanfuso and Neri (2024).

Figure 1: Belief updating across groups



Notes: The figure reports the average marginal effects of $E^i_{t-1}[x_{t+h}]$ for each socioeconomic characteristic k in (9), which identifies the average belief stickiness associated with that characteristic: $1 - \hat{G}_k = \hat{\beta}_1 + \hat{\beta}_{3,k} + \sum_{k' \neq k} \hat{\beta}_{3,k'} \bar{X}_{i,k',t}$, where $\bar{X}_{i,k',t} = \frac{1}{N} \sum_{i,t} X_{i,k',t}$. Table A.3 reports the detailed results. Sample: 2013M6-2023M5. Standard errors are clustered at the household and time level. Bars represent the 95% confidence interval.

Robustness A potential concern with our estimate of belief stickiness is that it may be driven by inexperienced consumers not understanding or paying attention to the survey questions. However, we find the opposite: belief stickiness is significantly higher among respondents with longer tenure in the survey and higher levels of numeracy. Another concern is that respondents may become fatigued over time and begin to mechanically repeat their previous answers. To address this, Figure A.2 replicates our analysis using only consumers who revise their forecasts between months t and $t-1$. The results remain unchanged, with belief stickiness still increasing in tenure, suggesting that respondent fatigue does not substantially drive the pattern.[14]

2.2 Belief stickiness over time

We now allow belief stickiness to vary over time and examine its fluctuations around the pandemic period, as well as its correlation with belief uncertainty.

[14]Figure A.18 confirms the robustness of our findings to an alternative calibration of prior expectations. In Figure A.12, we replicate the analysis using 1-year-ahead inflation expectations, and in Figure A.15, we measure expectations using the mean of consumers' density forecasts rather than point forecasts. In both cases, we find similar patterns for tenure, though the relationship with numeracy is weaker.

To do so, we classify consumers into groups based on the four socioeconomic characteristics that have the strongest impact on belief stickiness, as shown in Figure 1: tenure (in terciles), numeracy (high vs. low), education (college degree), and age (under 40). Each combination of these indicators defines a group, yielding a total of $J = 24$ distinct groups. We exclude group-month cells with fewer than 15 observations.

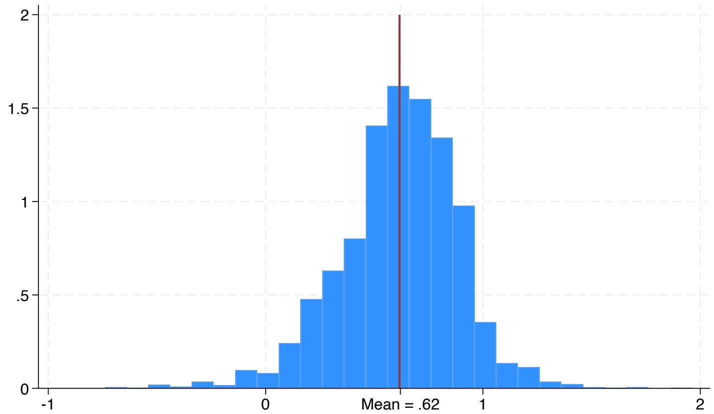
We estimate regression (7) separately for each group and month, obtaining a time-series of belief stickiness estimates $\hat{\beta}_{j,t} = 1 - G_{j,t}$. Figure 2 displays the distribution of these estimates. For each month t, we compute aggregate belief stickiness as the sample-size-weighted average across groups.

We find an average belief stickiness of $\overline{\widehat{1-G}} = 0.62$, which implies a new information gain of $\overline{\hat{G}} = 0.38$—that is, households place more weight on prior beliefs than on new information when forming beliefs. This level of rigidity is higher than estimates from surveys of professional forecasters (Coibion and Gorodnichenko, 2015a; Goldstein, 2023), but in line with strategic incentives-corrected estimates in **?**. This may reflect households' lower attention to new inflation-related news compared to professionals, or it may be driven by survey frequency: quarterly in professional forecasts versus monthly in the SCE. We note that our empirical approach does not speak to the optimality of belief stickiness—that is, whether the weight on new information G equals the Bayesian Kalman gain.[15] We return to the question of rationality in Section 3.4. Before that, we show that this average masks substantial fluctuations in belief stickiness over time.

Belief updating during the uncertainty surge Figure 3b shows a sharp decline in belief stickiness at the onset of COVID-19, indicating a surge in the responsiveness of inflation expectations to new information. Prior to March 2020, average stickiness is about 0.65;

[15] Some estimates of belief stickiness fall outside the rational expectations (RE) interval: approximately 3% of them are below zero, and 4% exceed one. While this may reflect measurement error, it could also stem from deviations from rational expectations. For instance, in the diagnostic expectations model of Bordalo et al. (2020), $G_t^j = (1+\theta)G_t^{REj}$, where $\theta > 0$ is a parameter governing overreaction to new information. In that case, our estimates becomes $\hat{\beta}_t^j = 1 - (1+\theta)G_t^{REj}$, which can fall below zero for sufficiently high θ and G_t^{REj}. Conversely, underreaction $\theta < 0$ combined with low G_t^{REj} could yield estimates above one.

Figure 2: Distribution of estimated belief stickiness



Notes: The figure plots the estimate of (7), weighted by the group-time size. The vertical line reports the weighted average. Sample period: 2013M1 - 2023M5.

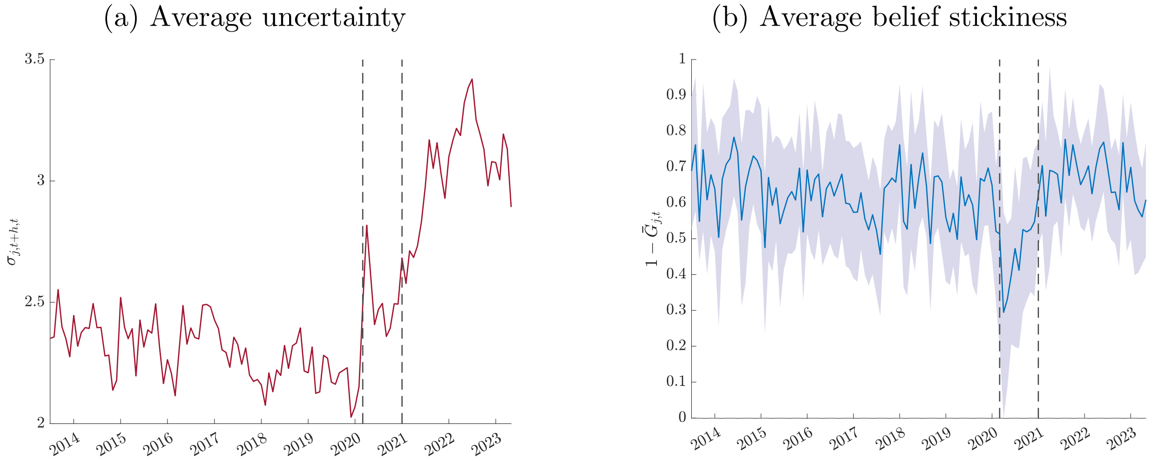
during the early pandemic it drops to roughly 0.3. Stickiness then mean-reverts and remains close to its pre-pandemic level through the subsequent high-inflation period. These dynamics are robust across alternative implementations: Appendix Figure A.13 shows the same pattern for the one-year horizon, and Appendix Figure A.6 obtains similar time-series dynamics when estimating stickiness on the full sample rather than within socioeconomic groups.[16]

One concern is that changes in estimated stickiness could reflect time variation in the inflation autoregressive parameter ρ used to map lagged forecasts into priors. Appendix Figure A.8 shows no evidence of a discrete change in ρ around 2020. As an external check that does not rely on ρ, Appendix Figure A.9 documents a contemporaneous decline in the share of Michigan Survey of Consumers respondents reporting that they heard no news about business conditions, consistent with a broad rise in attention at the onset of the pandemic.

Beyond levels, the pandemic marks a reversal in the co-movement of uncertainty and updating. During early 2020, posterior uncertainty rises sharply while stickiness falls (Figures 3a–3b), suggesting that households revise beliefs aggressively even as they become less certain about the outlook. From 2021 onward, uncertainty remains elevated while stickiness

[16]Estimating belief stickiness on the whole sample presents an advantage in terms of efficiency by increasing the sample size, but it could introduce a bias if the underlying belief stickiness actually differs across socioeconomic groups. Further details in Appendix H.

Figure 3: Belief stickiness pre- and post-pandemic



Notes: The left panel plots the average posterior belief uncertainty (standard deviation) across all consumers. The right panel plots the average belief stickiness across groups from regression 7, while the dashed blue lines represent the average 90% confidence interval. Standard errors are computed with a two-level bootstrap, within and across *(j,t)* groups. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

returns toward its pre-pandemic level.

These patterns complement recent evidence emphasizing the role of inflation levels in shaping household attention dynamics (Pfäuti, 2023; Weber et al., 2023). Our monthly estimates document an increase in expectation responsiveness to news before an increase in actual inflation, pointing towards uncertainty as a potential driver of attention dynamics.[17] The following reversal in the attention-uncertainty relationship suggests that attention may not depend solely on the level of uncertainty, but also on its source—a mechanism we investigate in Section 3. The implications for household financial decisions are discussed in Section 4.

[17]Supporting this interpretation, Figure A.9 shows a drop in the share of respondents in the Michigan Survey of Consumers who report not having heard any news about business conditions. This suggests a broad increase in consumer attention to economic conditions during the early pandemic.

3 Why uncertainty's source matters

In this section, we investigate the relationship between belief updating and belief uncertainty. We focus on the pre-2020 period and investigate how belief updating varies across households with different levels of prior and new information uncertainty. In the next section, we use these findings to structurally recover the average subjective prior and new information uncertainty and assess whether they can account for the observed decline in belief stickiness during the post-2020 period.

3.1 The implications of the Bayesian framework

Our empirical strategy estimates the monthly *updating gain* $G_{j,t}$ (or belief stickiness $1-G_{j,t}$), i.e. the weight households place on new inflation information when revising expectations. Our framework does not impose a specific belief formation model, but it nests the noisy information rational expectations (RE) model as a special case. This allows us to assess whether the observed properties of belief stickiness are consistent with standard Bayesian updating.

Under Bayes' rule, the gain equals the Kalman gain and is pinned down by relative precision:

$$1 - G_{j,t}^{RE} = \frac{\sigma_{e,j,t}^2}{\sigma_{e,j,t}^2 + \sigma_{j,t+h,t-1}^2} \tag{10}$$

where $\sigma_{j,t+h,t-1}^2$ is prior uncertainty and $\sigma_{e,j,t}^2$ is the variance of new-information noise (signal noise).

The RE model (10) yields two key testable implications. First, higher new information noise $\sigma_{e,t}^2$ is associated with higher belief stickiness. For example, households may face a higher cost of collecting information or a lower supply of information from newspapers, television, or social networks.

Second, higher prior uncertainty is associated with higher belief stickiness. The noisier the history of signals the agents received in the past, the more accurate the new signal will be *relative* to this stock of existing information. As a result, the agent assign more weight to the new signal in forming posterior beliefs, i.e. belief stickiness is lower.

We make two related observations. First, these implication holds similarly in models with endogenous information or rational inattention, where the signal noise $\sigma^2_{e,t}$ becomes a choice variable (Sims, 2003, 2006; Mackowiak and Wiederholt, 2009; Maćkowiak et al., 2023). Equation (10) shows that signal noise determines belief stickiness, regardless of whether it is endogenously chosen or externally determined. Second, while derived under the assumption of rational expectations, the comparative statics of equation (10) remain valid in many frameworks that build on the Bayesian benchmark but allow for behavioral deviations—such as diagnostic expectations (Bordalo et al., 2018, 2020; Bianchi et al., 2024a) and overconfidence (Broer and Kohlhas, 2024; Adam et al., 2025).[18]

3.2 Measuring prior uncertainty and noisy news in the SCE

Prior uncertainty We construct a proxy for prior belief uncertainty from the belief updating model in equation (4) and (5), which implies

$$\sigma^2_{i,j,t+h,t-1} = \rho^2 \sigma^2_{i,j,t+h-1,t-1} + \sigma^2_u \tag{11}$$

where $\sigma_{i,j,t+h-1,t-1}$ is last month's subjective uncertainty (from density forecasts) and ρ, σ_u are estimated from inflation data (Table A.1).

Our measure of prior uncertainty is valid only in the absence of structural changes in the volatility of the underlying inflation process σ_u. An increase in (real or perceived) fundamental volatility would raise households' prior uncertainty in month t for a given level

[18] In contrast, these results do not hold in models where the updating gain G_t is constant and independent of uncertainty, such as sticky information (Mankiw and Reis, 2002), adaptive learning with a constant gain (Orphanides and Williams, 2007; Eusepi and Preston, 2011), and the benchmark version of behavioral inattention (Gabaix, 2019).

of posterior uncertainty observed in the previous month $t-1$. The severe economic and social disruptions during the COVID period may have triggered such a structural change. For this reason, we restrict the analysis in this section to the pre-COVID period. In Section 4, we return to this issue and use our framework to provide evidence for such a structural break in fundamental volatility during COVID.

New information noise A main limitation of using natural variation in survey expectations compared to an experimental setting is that the econometrician does not directly observe the households' new information noise. To address this, we leverage our general framework to estimate it from the survey data.

We use our estimate of belief stickiness, together with our measures of prior and posterior uncertainty, to compute the new information noise from (5).[19]

$$\widehat{\sigma}^2_{e,j,t} = \frac{\sigma^2_{j,t+h,t} - (1-\hat{G}_{j,t})^2 \sigma^2_{j,t+h,t-1}}{\hat{G}^2_{j,t}}, \tag{12}$$

where $\hat{G}_{j,t} = 1 - \hat{\beta}_{j,t}$ is the estimated attention gain from (7), and $\sigma_{j,t+h,t}$ and $\sigma_{j,t+h,t-1}$ denote group averages of subjective (posterior and prior) uncertainty. We drop the observations $\widehat{\sigma}_{e,j,t}$ lower than zero, around 5% of the sample, and trim the top percentile to avoid outliers from measurement errors.

3.3 Opposite effects on belief updating

We now quantify how attention varies with the two uncertainty components in the pre-COVID sample.

We regress belief stickiness, our estimated object in Section 2, on our proxies for prior

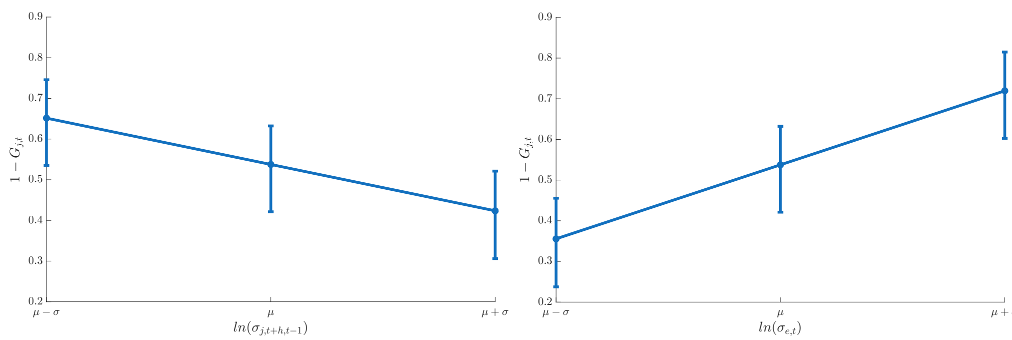
[19] We notice that equation (5) is part of our general framework and it does not impose any model of belief stickiness, meaning it does not make any assumption on G^j_t.

Table 1: Belief stickiness and uncertainty

	(1) $1 - G_{j,t}$	(2) $1 - G_{j,t}$	(3) $1 - G_{j,t}$	(4) $1 - G_{j,t}$
$ln(\sigma_{j,t+h,t-1})$	-0.198***		-0.410***	-0.263***
	(0.034)		(0.029)	(0.048)
$ln(\widehat{\sigma}_{e,j,t})$		0.301***	0.359***	0.355***
		(0.021)	(0.027)	(0.029)
Year-Month FEs	Y	Y	Y	Y
Group FE	N	N	N	Y
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Adjusted R-squared	0.12	0.47	0.63	0.65
Observations	768	768	768	768

Notes: This table reports the estimated coefficient from regression (13). Standard errors (in parentheses) are computed with a two-level bootstrap, within and across *(j,t)* groups. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

Figure 4: Belief stickiness and uncertainty



Notes: The figure represents graphically the estimated coefficients from column (4) of Table 1. It shows the relationship between belief stickiness and prior uncertainty (left panel) and new information noise (right panel).

uncertainty and new information noise:

$$1 - G_{j,t} = \alpha + \delta_1 ln(\Sigma_{j,t+h,t-1}) + \delta_2 ln(\widehat{\sigma}_{e,j,t}) + \gamma_t + \gamma_j + \epsilon_{j,t}, \tag{13}$$

with time and group-level fixed effects. To account for the errors related to the generated regressor $\widehat{\sigma}_{e,j,t}$, we bootstrap the standard errors with 10000 draws both within time-groups (regression 7) and across time-groups (regression 13).

We test two hypotheses. First, for a given new information noise, higher prior uncertainty leads to lower belief stickiness: $\delta_1 < 0$. Second, for a given prior uncertainty, higher new information noise leads to lower belief stickiness: $\delta_2 > 0$.

Our results, reported in Table 1, strongly support both predictions. Column (4) shows that new information noise is positively associated with belief stickiness, while the effect of prior uncertainty is negative. Figure 4 illustrates these effects: a one-standard-deviation increase in prior uncertainty reduces belief stickiness by about 0.1 ($\approx 20\%$), while a similar increase in new information noise raises it by about 0.15 ($\approx 30\%$). In short, belief stickiness is robustly negatively related to prior uncertainty and positively related to new information noise, consistent with the Bayesian framework.

Discussion Because $\widehat{\sigma}_{e,j,t}$ is constructed using $\hat{G}_{j,t}$, measurement error could introduce a bias in regression (13). Three observations mitigate this concern. First, we removed the top percentile of $\widehat{\sigma}^2_{e,j,t}$ to limit the influence of outliers. Second, measurement error does not affect our proxy for prior uncertainty $\sigma_{i,j,t+h,t-1}$, , which still shows the expected negative effect in Column (1). Third, in Appendix G, we replicate the analysis using two alternative proxies for new information noise—posterior uncertainty and realized forecast errors (controlling for prior uncertainty)—and obtain qualitatively similar results. While these alternatives cannot perfectly isolate new information noise, they reinforce the robustness of our main findings.[20]

A distinct class of explanations emphasizes the level of inflation as a driver of attention (Pfäuti, 2023; Weber et al., 2025). To assess this possibility, Table A.8 reports our estimates when controlling for annual, monthly, and lagged monthly inflation. We find no significant effect of inflation on belief stickiness in the pre-COVID sample. This is consistent with the idea that the information environment—captured by the source of uncertainty—helps explain attention regimes beyond the level of inflation itself.

[20] Recent studies suggest that households' belief updating may depend on the level of inflation (Pfäuti, 2023; Weber et al., 2025). To assess this possibility, Table A.8 reports our estimates when controlling for annual, monthly, and lagged monthly inflation. We find no significant effect of inflation on belief stickiness in the pre-COVID sample.

3.4 Overreaction relative to the Bayesian benchmark

The results above support the *sign* predictions of the Bayesian noisy-information benchmark: attention rises with prior uncertainty and falls with noisier news. A natural next question is whether households are also *quantitatively* Bayesian—that is, whether the *level* of attention we estimate in the data coincides with the Kalman gain implied by the uncertainty components we recover.

A rational-expectations counterfactual. In the Bayesian benchmark, the optimal gain is pinned down by relative precision (equation (10)). Using our pre-COVID proxies for prior uncertainty $\sigma^2_{j,t+h,t-1}$ and new-information noise $\widehat{\sigma}^2_{e,j,t}$ (equations (11) and (12)), we can construct the *counterfactual* rational-expectations (RE) stickiness:

$$1 - \tilde{G}^{RE}_{j,t} = \frac{\widehat{\sigma}^{j,2}_{e,t}}{\widehat{\sigma}^{j,2}_{e,t} + \sigma^2_{j,t+h,t-1}}. \tag{14}$$

Under quantitative Bayesian updating, we should have $\widehat{1 - G_{j,t}} \approx 1 - \widetilde{G}^{RE}_{j,t}$.[21]

Figure 5 reports the results for each group-month in the pre-COVID sample.[22]

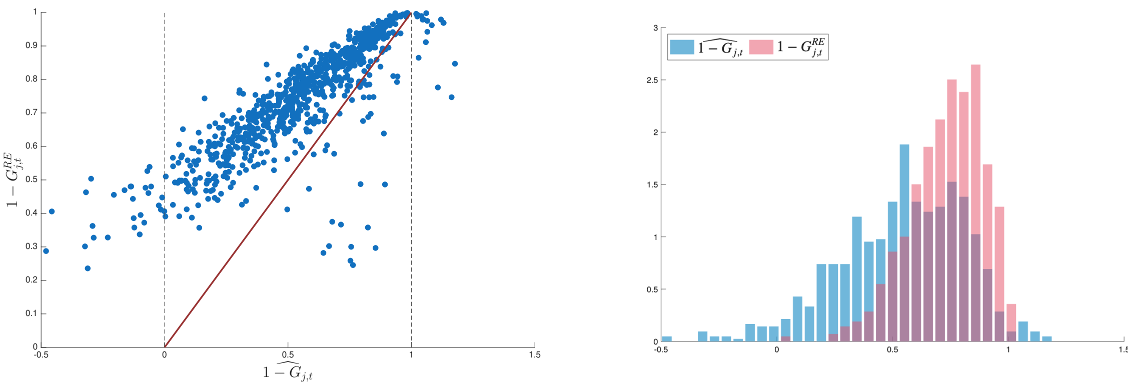
We find that households consistently display lower belief stickiness than predicted by the RE benchmark. In Figure 5a, most estimates lie above the 45-degree line, indicating that households assign more weight on new information than the rational counterfactual. Similarly, Figure 5b shows that the distribution of estimated stickiness centers around 0.6, compared with about 0.75 under RE. In other words, households in the SCE assign roughly 25% more weight to new information than the rational benchmark would imply, systemati-

[21]Because $\widehat{\sigma}^2_{e,j,t}$ is recovered from *subjective* uncertainty, it reflects *perceived* information noise. Therefore, equality with (14) is a necessary but not sufficient condition for full rationality: it requires that households apply Bayes' rule given their perceived signal quality, but it does not impose that perceived and actual signal precision coincide.

[22]Because we extract new information noise from individual subjective uncertainty, our measure reflects *perceived* information noise, which may differ from the *actual* noise. Thus, a belief stickiness equal to (14) is a necessary but not sufficient condition for rationality. Full rationality would also require that perceived and actual information accuracy coincide. We show that even this necessary condition is rejected in the data.

Figure 5: Counterfactual RE and estimated belief stickiness

(a) Comparison (b) Distributions



Notes: The left panel plots on the x-axis the estimated belief stickiness $\hat{\beta}_{j,t} = \widehat{1-G_{j,t}}$ from regression (7) and on the y-axis the implied rational expectation belief stickiness $1-G^{RE}_{j,t}$ from equation (10). If the observation lies above the 45-degree red line, then the implied RE belief stickiness is larger than the estimated one. The right panel plots the two distributions, showing that the distribution of the implied RE belief rigidities shifted to the right of the estimated ones. The difference between the two means is ≈ 0.18, with standard error ≈ 0.007 (two-level bootstrap, within and across *(j,t)* groups). Sample Jun13-Feb20.

cally overreacting relative to the Rational Expectations framework. [23]

Estimating a belief formation model Next, we investigate potential biases in consumer belief formation by testing the following functional form for estimated belief stickiness:

$$1 - G_{j,t} = 1 - \left(\bar{g} + (1+\theta) \frac{\sigma^2_{j,t+h,t-1}}{(1-\alpha)\widehat{\sigma}^{j,2}_{e,t} + \sigma^2_{j,t+h,t-1}} \right) \tag{15}$$

We introduce three parameters that can generate a wedge between the estimated belief stickiness and the RE benchmark in (14): $\bar{g}$, θ, and α, each capturing different behavioral departures. If all are zero, the model collapses to the Rational Expectations Kalman gain. First, $\alpha \neq 0$ captures a relative mis-weighting of new information noise versus prior uncertainty compared to G^{RE}. When $0 < \alpha < 1$, this bias resembles overconfidence (Broer and Kohlhas, 2024; Adam et al., 2025).[24] Second, $\theta \neq 0$ (with $\alpha = 0$) represents a proportional

[23] Figure A.3 replicates the comparison excluding respondents in the lowest tercile of survey tenure, i.e., considering only consumers participating in the survey for longer periods. The results are virtually unchanged.

[24] An important difference between the bias we test here and overconfidence is that in the latter framework agents overweight new information because they underestimate its noise. Instead, we extract the new

Table 2: GMM estimation

	(1)	(2)	(3)	(4)
θ	0.642***			0.306**
	(0.034)			(0.128)
α		0.619***		0.289*
		(0.010)		(0.162)
$\bar{g}$			0.185***	0.027
			(0.007)	(0.018)
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Observations	768	768	768	768

Notes: This table reports the estimated coefficients θ, α and $\bar{g}$ from objective function (15) with GMM, weighting by *(j,t)* group size. $\alpha \neq 0$ and $\theta \neq 0$ capture different forms of overreaction documented in the literature, and $\bar{g}$ allows for a constant component of belief stickiness. Standard errors (in parentheses) are computed with a two-level bootstrap, within and across *(j,t)* groups. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

distortion of the Kalman gain, akin to diagnostic expectations (Bordalo et al., 2018, 2020).[25] Third, $\bar{g}$ allows for a constant component of stickiness unrelated to information accuracy. We estimate the three parameters by generalized method of moments, minimizing the distance between (15) and our empirical stickiness estimates. Table 2 reports the results.

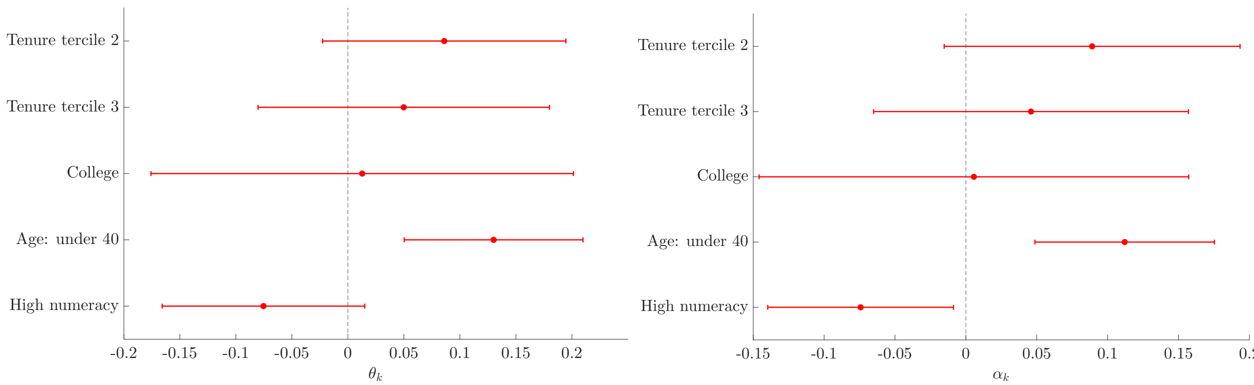
We find $\alpha > 0$ and $\theta > 0$, while $\bar{g}$ is not significantly different from zero. Both results imply that households overweight new relative to prior information. Thus, while households are *qualitatively* Bayesian — updating more when prior beliefs are uncertain and less when signals are noisy — they are not *quantitatively* Bayesian. Instead, they consistently place too much weight on new information relative to the rational benchmark, leading to systematic overreaction.

Heterogeneity in overreaction bias We investigate whether the higher updating gain of older and high-numeracy households (Section 2.1) reflects better information or a smaller behavioral bias. We interact the overreaction parameters in equation (15) with socioeco-

information noise from consumers' subjective beliefs. Therefore, we estimate how much they overweigh their subjective new information noise.

[25]An important difference between the bias we test here and diagnostic expectation is that in the latter framework the bias only applies to the first moment of belief and not to the second moment. In our framework, the same belief stickiness is applied to first and second moments of beliefs, equation (4) and (5) (Bordalo et al., 2020). Bianchi et al. (2024b) proposes a "smooth" DE setting where the bias also affects the second moment of beliefs.

Figure 6: Heterogeneity in overreaction bias



(a) Heterogeneity in θ

(b) Heterogeneity in α

Notes: The left panel plots the estimated interactions with socioeconomic dummies and θ in 15. Table A.7 reports the detailed results. The right panel plots the estimated interactions with socioeconomic dummies and α in 15. Table A.6 reports the detailed results. Confidence intervals are at the 90% level and standard errors are computed with a two-level bootstrap, within and across *(j,t)* groups. Sample Jun13-Feb20.

nomic dummies, estimating $\theta = \theta_0 + \sum_{k=1}^{5} \theta_k D_k$, where D_k indicates membership in group k. Figure 6a and Table A.7 report the results. Younger consumers display a significantly higher overreaction bias, while high-numeracy consumers display a lower one. Interacting socioeconomic dummies with α yields similar patterns (Figure 6b). These results indicate that the lower updating gain of older and high-numeracy households stems from a smaller overreaction bias rather than from lower information quality. The intuitive direction of these heterogeneities also reduces concerns that our findings are driven by measurement error.

4 Belief stickiness as a diagnostic tool

Section 3 showed that prior uncertainty and signal noise have opposite effects on attention: when households are less confident in their priors they update more, while noisier news makes them update less. This sign pattern signifies that attention is also a diagnostic. Joint movements in (i) subjective inflation uncertainty and (ii) attention identify whether uncertainty is driven primarily by changes in perceived fundamentals (priors becoming obsolete) or by a deterioration in information quality (noisier signals). We now use this logic to interpret the post-2020 period.

4.1 Unpacking the post-2020 uncertainty surge

In the post-2020 period, structural changes in macroeconomic volatility can invalidate the new information noise proxy we derived in Section 3.2. Instead, we recover the two underlying sources of uncertainty directly from observables using two relationships: (i) the second-moment identity implied by linear updating, equation (5), and (ii) the belief updating equation (15), estimated on pre-COVID data.[26]

Solving the resulting system yields expressions for $\sigma^2_{e,j,t}$ (noisy news) and $\sigma^2_{j,t+h,t-1}$ (prior uncertainty) as functions of the observables $\widehat{G}_{j,t}$ and $\sigma_{j,t+h,t}$, and of the pre-COVID parameters $(\hat{\alpha}, \hat{\theta})$:

$$\begin{aligned} \sigma^2_{e,j,t} &= \frac{\big(\theta + (1 - G_{j,t})\big)\, \sigma^2_{j,t+h,t}}{(1 - G_{j,t})^2 (1-\alpha) G_{j,t} + G^2_{j,t}\big(\theta + (1 - G_{j,t})\big)}, \\ \sigma^2_{j,t+h,t-1} &= \frac{(1-\alpha) G_{j,t}}{\theta + (1 - G_{j,t})}\, \sigma^2_{e,j,t}. \end{aligned} \tag{16}$$

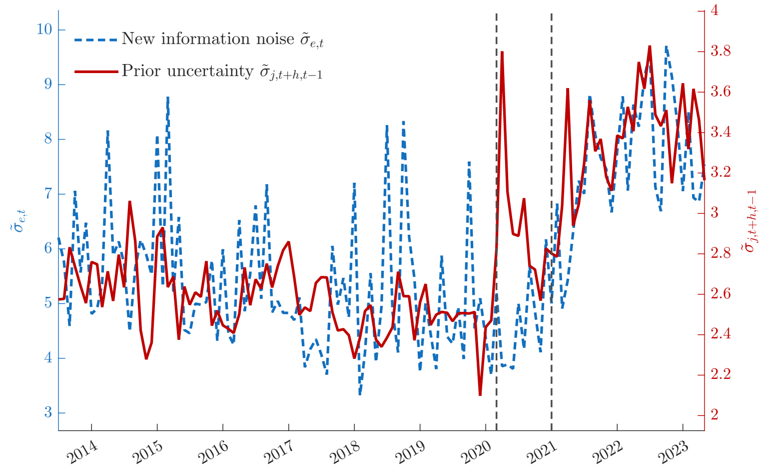
We apply (16) to each (j, t) cell and then aggregate using group-size weights. Figure 7 reveals a clear regime shift in the composition of subjective inflation uncertainty.

Post-COVID From 2021, after the initial months of COVID-19, we document a sharp rise in new information noise (Figure 7). The increase in noise *relative* to prior uncertainty accounts for the positive correlation between belief uncertainty and stickiness observed in Figure 3. Later, although new information noise remains elevated, its relative weight compared to prior uncertainty returns to pre-COVID levels. This shift helps explain why belief stickiness resembles pre-COVID values, even as overall belief uncertainty is higher in the post-COVID period.

Higher information noise may result from consumers facing greater costs in gathering

[26] If the behavioral mapping between attention and relative precision shifted in the high-inflation regime, the implied levels would change, but the qualitative diagnostic based on the opposite comovement patterns remains. Appendix Figure A.4 shows that the inferred regime shift is similar under the Bayesian benchmark ($\alpha = \theta = 0$).

Figure 7: Decomposing uncertainty: prior and new information noise

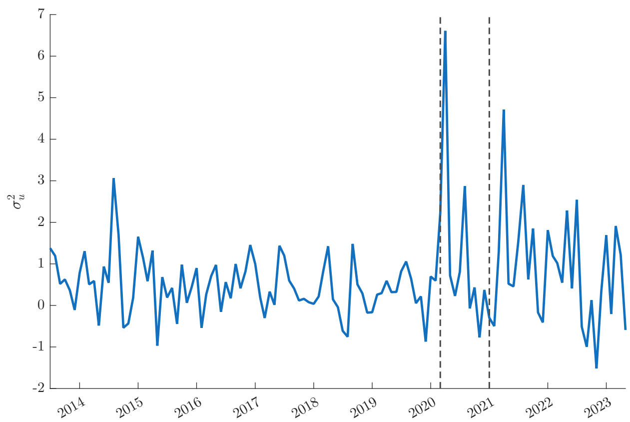


Notes: The figure plots the estimates of new information noise (left y-axis) and prior uncertainty (right y-axis) from equation (16), with the estimates of estimates of belief stickiness $\widehat{1-G_{j,t}}$, posterior uncertainty $\sigma_{j,t+h,t}$ and $\hat{\alpha}, \hat{\theta}$.

information or a reduced supply of information from newspapers, television, and social networks. This period, beginning in February 2021, is also marked by rising inflation and increased media coverage. At first glance, this may seem inconsistent with the observed increase in noise in new information about inflation. However, greater media coverage does not necessarily imply more accurate information dissemination. Instead, it may reflect efforts to manage heightened uncertainty driven by the increasing difficulty of predicting inflation, in a moment when even economists disagreed on how long the high-inflation period would have lasted (Markovitz, 2021).

COVID outbreak Interestingly, we find a sharp increase in prior uncertainty at the COVID-19 outbreak, in March 2020. This follows from the contemporaneous spike in belief uncertainty and decline in belief stickiness displayed in Figure 3. This might reveal a structural change in the underlying volatility of the economy in the post-COVID era, which can be traced to two possible sources.

Figure 8: Model-implied fundamental volatility



Notes: The figure plots the estimates of fundamental volatility $\sigma^2_{u,t} = \tilde{\sigma}^2_{t+h,t-1} - \rho^2 \sigma^2_{t+h-1,t-1}$.

4.2 Prior uncertainty from macroeconomic volatility

First, consider an increase in fundamental volatility ${\sigma'_u}^2 > {\sigma_u}^2$. As shown in equation (11), such higher volatility implies that prior information becomes obsolete, and therefore more uncertain, when forecasting the future, as the stochastic process of the fundamental becomes more unpredictable. We do not take a stand on what could have driven such an increase in fundamental volatility, as some determinants might have been specific to the COVID-19 case: the lethality of the virus, the capacity of healthcare systems to meet an extraordinary challenge, its economic consequences, the waiting time to develop a safe vaccine, et cetera (see Baker et al. (2020)).

We recover the level of fundamental volatility implied by our measure of prior uncertainty by inverting equation (11). Specifically, we compute $\sigma^2_{u,t} = \tilde{\sigma}^2_{t+h,t-1} - \rho^2 \sigma^2_{t+h-1,t-1}$, where $\tilde{\sigma}_{t+h,t-1} \equiv \int^j \tilde{\sigma}_{j,t+h,t-1} dj$ and $\sigma_{j,t+h-1,t-1} \equiv \int^j \sigma_{j,t+h-1,t-1} dj$. Figure 8 shows that fundamental volatility is broadly stable in the pre-COVID period but rises sharply at the onset of the pandemic, driving the observed increase in prior uncertainty.

Second, consider a loss of trust in the central bank's ability to maintain its inflation target. In this stylized setting, one can think of the belief about the long-run mean of inflation becoming uncertain, $\mu_x \sim N(\bar{\pi}, \sigma_\mu^2)$. Prior uncertainty then becomes

$$\sigma^2_{j,t+h,t-1} = (1-\rho)^2\sigma^2_\mu + \rho^2\sigma^2_{j,t+h-1,t-1} + \sigma^2_u. \tag{17}$$

A larger uncertainty about long-run mean inflation σ^2_μ would increase prior uncertainty and, as a result, lower belief stickiness. This could be possibly due to lower trust in the central bank's ability to achieve its price stability objective. Aikman et al. (2024) measures the trust in the Federal Reserve from Twitter data and finds the lowest point at the COVID outbreak. Alternatively, it could be more generally due to uncertainty about economic policy implementation.[27]

A regime shift in the autoregressive parameter ρ could also have played a role. However, detecting a structural change within such a short sample is challenging. In Appendix I, we present rolling-window AR(1) regressions of monthly inflation and show that there is no strong evidence for a shift in the autoregressive parameter ρ around the COVID-19 period compared to previous periods.

Another possible force driving the change in belief stickiness during COVID-19 is the lockdown policy restrictions on movements implemented by policymakers to stop the spread of the virus. These restrictions might have lowered the cost of browsing for news and therefore contributed to this decrease in belief stickiness. We investigate this in the next section.

5 Conclusion

Uncertainty does not have a uniform effect on expectations; its consequences depend on its source. When uncertainty reflects less confidence in prior beliefs, households place greater weight on incoming information and update more. When uncertainty reflects noisier signals,

[27] Figure A.5 plots the Economic Uncertainty Index, constructed by Baker et al. (2016) and offers additional evidence for this implication. We plot the "news coverage component", which measures the share of articles in US online newspapers that mention economic policy uncertainty. This index spikes up during the Covid outbreaks and decreases to the pre-Covid level in the following period, corroborating our results that the increase in belief uncertainty observed during Covid might be due to this policy uncertainty.

updating diminishes and prior beliefs exert greater influence. This framework organizes the post-2020 evidence: the early pandemic is consistent with a shift in fundamentals that raised prior uncertainty and accelerated updating, whereas the subsequent high-inflation phase features rising uncertainty alongside greater stickiness, indicative of noisier information. The estimates accord with Bayesian comparative statics while revealing systematic overreaction in levels relative to a rational benchmark.

Our results have important policy-relevant implications. Because belief stickiness depends on the source of uncertainty, it can be used to identify whether it is driven by noisier signals —where clearer communication and information provision can help—or elevated fundamental volatility, where communication has limited traction and broader stabilization tools are required. With high-frequency expectations data, tracking stickiness offers a practical, real-time read on the uncertainty source to inform policy.

Our work also suggests several directions for future research. While this paper focuses on household expectations, understanding the determinants of information frictions on the production side of the economy is equally important to improve macroeconomic models, allowing for a more precise study of macroprudential policy and its dependence on different sources of uncertainty. In addition, while this paper studies the total adjustment of beliefs, future work can use this methodology to distinguish between intensive and extensive margins of belief adjustment. Finally, belief stickiness could be used empirically to examine the macroeconomic effects of different types of uncertainty.

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Online Appendix

A Belief formation models

The framework in equation (4) encompasses not only Rational Expectations but also alternative models of belief formation, particularly those that incorporate frictions in processing new information. Some examples are:

- Rational expectations: $G_t^{RE} = \frac{\tau_t}{\tau_t + \Sigma_{t+h,t-1}^{-1}}$, where $\Sigma_{t+h,t-1} \equiv var(x_{t+h} - E_{t-1}^i[x_{t+h}])$ is the prior variance (Sims, 2003; Woodford, 2001; Mackowiak and Wiederholt, 2009). In the case of full-information, the signal is perfectly informative, $\tau_t \to \infty$, and therefore $G_t = 1$.

- Diagnostic expectation (Bordalo et al., 2018, 2020): households overreact to new information according to $\theta > 0$, therefore $G_t = (1+\theta)G_t^{RE}$.

- Overconfidence (Broer and Kohlhas, 2024; Adam et al., 2025): households perceived signal accuracy as more accurate, $\tilde{\tau}_t > \tau_t$, and therefore $G_t = \frac{\tilde{\tau}_t}{\tilde{\tau}_t + \Sigma_{t+h,t-1}^{-1}} > G_t^{RE}$.

- Sticky information (Mankiw and Reis, 2002): household has a probability $1-\lambda$ of fully updating her beliefs $G_t = 1$, and λ of not updating their belief at all, $G_t = 0$.

- Memory frictions (Azeredo da Silveira et al., 2024): agents pay a cost not only to process new information but also past information, meaning that prior accuracy decay with time. Then, G_t would internalize both the noise of new information and the memory friction.

- Learning with constant gain (Eusepi and Preston, 2011): households learn about the model's parameters in each period using a constant gain, so that they never learn completely.

- Behavioral inattention (Gabaix, 2019): agents pay limited attention to some variable x for non-rational reasons. It can map in our framework if $G_t = m_t$ and the "default" belief is the prior.

By contrast, the structure in (4) is not consistent with belief models in which agents lack knowledge of the true economic structure and instead rely on alternative perceived laws of motion or simple heuristics, such as natural expectations (Fuster et al., 2010), over- or under-extrapolation (Angeletos et al., 2021), or narrative-based beliefs (Andre et al., 2024).

B Point estimates and subjective distribution of inflation in the SCE

Q9c

And in your view, what would you say is the percent chance that, **over the 12-month period between August 2015 and August 2016 …**

Instruction H4.

the rate of inflation will be 12% or higher	____ percent chance
the rate of inflation will be between 8% and 12%	____ percent chance
the rate of inflation will be between 4% and 8%	____ percent chance
the rate of inflation will be between 2% and 4%	____ percent chance
the rate of inflation will be between 0% and 2%	____ percent chance
the rate of deflation (opposite of inflation) will be between 0% and 2%	____ percent chance
the rate of deflation (opposite of inflation) will be between 2% and 4%	____ percent chance
the rate of deflation (opposite of inflation) will be between 4% and 8%	____ percent chance
the rate of deflation (opposite of inflation) will be between 8% and 12%	____ percent chance
the rate of deflation (opposite of inflation) will be 12% or higher	____ percent chance
Total	**100**

C Alternative measure of belief uncertainty

A drawback of using bins questions to measure the individual density forecast is that the intervals considered in the bins might too wide or too narrow to capture the whole belief distribution fully. The bins of the Survey of Consumer Expectations range from -12% to 12% in steps of 2 to 4 percentage points. During high inflation periods, such as the post-Covid months, consumers might attribute a large probability on inflation realization above this upper bound, which could lead to inaccuracies in measuring their true belief distribution from bins question.

To address this concern, we compare our benchmark measure with an alternative measure of belief uncertainty based on the rounding of point forecasts, as in Binder (2017). In each month we compute the share of respondents that provide a forecast multiple of 5. This uncertainty measure is based on the Round Numbers Suggest Round Interpretation (RNRI) principle, which suggests that round numbers are frequently used to convey that a quantitative expression should be interpreted as imprecise (Krifka, 2007).

Figure A.1 plots the measure based on rounding ('Share M5', from Binder (2017)) together with the average uncertainty measures from the bins survey question: the standard

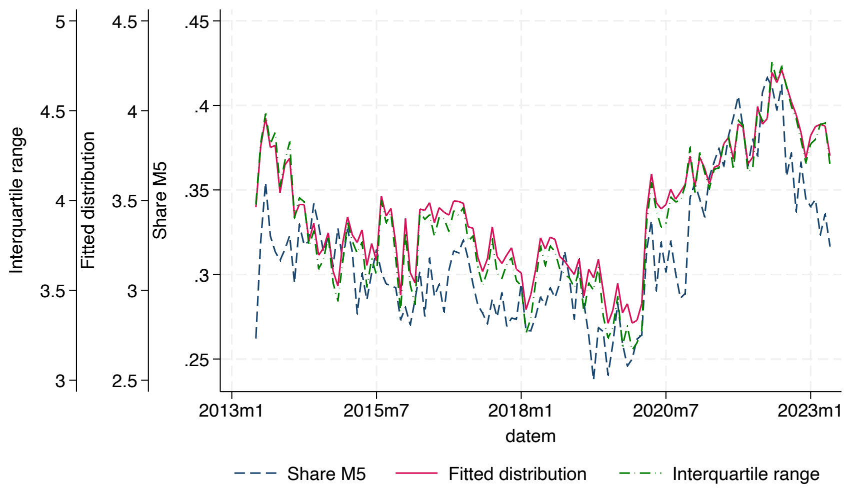


Figure A.1: Uncertainty measures

Notes: This plot shows different measures of subjective uncertainty over time. The blue dashed line is the share of respondents that provide a forecast multiple of 5 from Binder (2017), the red solid line is the average subjective standard deviation provided by the SCE, and the green dotted line is the average subjective interquartile range provided by the SCE. Sample period: 2013M6-2023M5. Bars represent the 95% confidence interval.

deviation of the fitted distribution and the interquartile range. The rounding uncertainty measure closely tracks the other two and exhibits the same pattern during and after the COVID period.

D Inflation autoregressive process

We estimate (1) with the following regression on the 12-month inflation rate x_t:

$$x_t = (1 - \rho)\mu_x + \rho x_{t-1} + u_t, \tag{A.1}$$

and we recover the long-run mean $\mu_x = \frac{\hat{\alpha}}{(1-\hat{\rho})}$ and the standard deviation of the fundamental noise $\sigma_u = std(\hat{u}_t)$. We run the regression on different subsamples, and the resulting coefficients are quite stable.

E Additional tables

Table A.1: Inflation autoregressive proces

	(1) $Inflation$	(2) $Inflation$	(3) $Inflation$	(4) $Inflation$
L.$Inflation$	0.946***	0.975***	0.968***	0.934***
	(0.016)	(0.013)	(0.007)	(0.023)
μ_x	2.3	2.82	2.47	2.15
σ_u	.38	.4	.38	.44
Sample	Jan90-Feb20	Jan90-May23	Jan80-Feb20	Jan00-Feb20
Observations	361	394	481	241

Notes: $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

Table A.2: Descriptive Statistics

	Mean	SD	Min	Max	N
Beliefs					
$Forecast$	5.24	8.68	-50	75	131007
$Revision$	-0.21	7.27	-100	110	95098
$Post\ Uncert$	2.94	3.06	0	22	133362
$Post\ Uncert\ IQR$	3.32	3.52	0	28	133362
Socioeconomic characteristics					
$College_{it}$	0.89	0.31	0	1	135669
$Income\ 50kto100k_{it}$	0.35	0.48	0	1	134293
$Income\ Over100k_{it}$	0.30	0.46	0	1	134293
$Income\ Under50k_{it}$	0.34	0.47	0	1	134293
$High\ Numeracy_{it}$	0.74	0.44	0	1	135610
$Female_i$	0.47	0.50	0	1	135606
Age_{it}	50.57	15.25	17	94	135549
$White_i$	0.85	0.35	0	1	135663
$Tenure_{it}$	5.62	3.39	1	16	135669

Notes: This table provides descriptive statistics for beliefs and household socioeconomic characteristics derived from the Survey of Consumer Expectations (SCE). The sample period is 2013M6-2023M5.

Table A.3: Heterogeneity in Belief Updating

	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$	(4) $E_t^i[x_{t+h}]$	(5) $E_t^i[x_{t+h}]$	(6) $E_t^i[x_{t+h}]$	(7) $E_t^i[x_{t+h}]$	(8) $E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.573***	0.455***	0.526***	0.603***	0.561***	0.524***	0.557***	0.396***
	(0.011)	(0.017)	(0.023)	(0.015)	(0.019)	(0.015)	(0.024)	(0.041)
$Tenure\ Tercile{=}2 \times E_{i,t-1}[x_{t+h}]$		0.120***						0.116***
		(0.021)						(0.020)
$Tenure\ Tercile{=}3 \times E_{i,t-1}[x_{t+h}]$		0.234***						0.223***
		(0.020)						(0.021)
$College_{it}\ {=}1 \times E_{i,t-1}[x_{t+h}]$			0.053**					0.034
			(0.025)					(0.025)
$Age\ Over60{=}1 \times E_{i,t-1}[x_{t+h}]$				-0.023				-0.038*
				(0.022)				(0.023)
$Age\ Under40{=}1 \times E_{i,t-1}[x_{t+h}]$				-0.085***				-0.079***
				(0.025)				(0.024)
$Income\ Over100k{=}1 \times E_{i,t-1}[x_{t+h}]$					-0.011			-0.020
					(0.033)			(0.032)
$Income\ Under50k{=}1 \times E_{i,t-1}[x_{t+h}]$					-0.003			0.015
					(0.022)			(0.021)
$High\ Numeracy{=}1 \times E_{i,t-1}[x_{t+h}]$						0.075***		0.060***
						(0.018)		(0.018)
$Female{=}1 \times E_{i,t-1}[x_{t+h}]$							-0.025	-0.009
							(0.022)	(0.022)
$White{=}1 \times E_{i,t-1}[x_{t+h}]$							0.033	0.018
							(0.022)	(0.022)
Constant	2.038***	1.975***	2.048***	2.038***	2.114***	2.042***	2.050***	2.068***
	(0.053)	(0.051)	(0.053)	(0.053)	(0.055)	(0.050)	(0.054)	(0.051)
Year-Month FEs	Y	Y	Y	Y	Y	Y	Y	Y
Non-interacted variables	N	Y	Y	Y	Y	Y	Y	Y
Year-Month × variables	N	Y	Y	Y	Y	Y	Y	Y
Sample	Jun13-May23	Jun13-May23	Jun13-May23	Jun13-May23	Jun13-May23	Jun13-May23	Jun13-May23	Jun13-May23
Adjusted R-squared	0.33	0.33	0.33	0.33	0.33	0.34	0.33	0.35
Observations	95098	95098	95098	95062	94164	95070	95059	94101

Notes: Standard errors (in parentheses) are clustered at the individual and time levels. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

Table A.4: Descriptive Statistics

	Mean	SD	Min	Max	N
Lockdown policies					
School	1.45	0.96	0	3	35859
Workplace	0.82	0.91	0	3	35859
Event	0.72	0.79	0	2	35859
Gathering	1.44	1.78	0	4	35859
Transport	0.25	0.47	0	2	35859
StayAtHome	0.48	0.67	0	2	35859
Movements	0.45	0.66	0	2	35859
Travel	0.24	0.58	0	2	35859
CasesCOVID	0.01	0.01	0	0.103	35859
DeathsCOVID	0.00	0.00	0	0.00108	35859
Economic Policy Uncertainty					
EPUState	1.98	1.88	0	14.66	40756
EPUNational	1.97	1.53	0	15.63	40756
EPUComposite	3.23	2.47	0.151	19.64	40756

Notes: This table provides descriptive statistics for lockdown policy intensity (from Hale et al. (2020)) and economic policy uncertainty (from Baker et al. (2022)). The sample period is 2020M3-2023M5.

Table A.5: Belief stickiness and lockdown measures

	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$	(4) $E_t^i[x_{t+h}]$	(5) $E_t^i[x_{t+h}]$	(6) $E_t^i[x_{t+h}]$	(7) $E_t^i[x_{t+h}]$	(8) $E_t^i[x_{t+h}]$	(9) $E_t^i[x_{t+h}]$	(10) $E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.586***	0.616***	0.633***	0.610***	0.559***	0.626***	0.565***	0.548***	0.597***	0.655***
	(0.130)	(0.113)	(0.119)	(0.116)	(0.122)	(0.112)	(0.121)	(0.124)	(0.110)	(0.114)
$E_{i,t-1}[x_{t+h}] \times ln(DeathsCOVID)$	-0.030*	-0.020	-0.019	-0.025	-0.033*	-0.020	-0.032*	-0.036*	-0.020	-0.016
	(0.016)	(0.016)	(0.017)	(0.017)	(0.018)	(0.015)	(0.018)	(0.019)	(0.015)	(0.016)
$E_{i,t-1}[x_{t+h}] \times ln(CasesCOVID)$	0.046**	0.033*	0.038**	0.046**	0.052**	0.038**	0.051**	0.057**	0.032*	0.034*
	(0.019)	(0.017)	(0.018)	(0.019)	(0.020)	(0.015)	(0.020)	(0.021)	(0.016)	(0.017)
$E_{i,t-1}[x_{t+h}] \times School$	-0.036*								0.009	
	(0.019)								(0.025)	
$E_{i,t-1}[x_{t+h}] \times Workplace$		-0.068***							-0.069**	
		(0.023)							(0.034)	
$E_{i,t-1}[x_{t+h}] \times Event$			-0.063**						-0.001	
			(0.027)						(0.040)	
$E_{i,t-1}[x_{t+h}] \times Gathering$				-0.023**					0.020	
				(0.011)					(0.015)	
$E_{i,t-1}[x_{t+h}] \times Transport$					-0.079**				-0.032	
					(0.038)				(0.038)	
$E_{i,t-1}[x_{t+h}] \times StayAtHome$						-0.081***			-0.050	
						(0.029)			(0.042)	
$E_{i,t-1}[x_{t+h}] \times Movements$							-0.053**		0.003	
							(0.024)		(0.038)	
$E_{i,t-1}[x_{t+h}] \times Travel$								-0.055	-0.018	
								(0.037)	(0.042)	
$E_{i,t-1}[x_{t+h}] \times Lockdown$										-0.084***
										(0.031)
Constant	2.240***	2.252***	2.255***	2.250***	2.252***	2.248***	2.252***	2.244***	2.248***	2.256***
	(0.086)	(0.079)	(0.082)	(0.084)	(0.087)	(0.080)	(0.086)	(0.089)	(0.077)	(0.081)
Year-Month FEs	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year-Month × Variables	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Non-interacted variables	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Sample	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22
Adjusted R-squared	0.35	0.34	0.34	0.34	0.34	0.34	0.34	0.34	0.34	0.34
Observations	24845	25626	25626	25626	25626	25626	25626	25626	25626	25626

Notes: Variables *School* to $Travel$ measure lockdown policies intensity for different social activities, from the Oxford Covid-19 Government Response Tracker (OxCGRT). *Lockdown* is the average of the other lockdown indicators. We control for year-month fixed effects, and for socioeconomic characteristics, such as education, income, age, gender, race, and tenure. Covid Controls are $ln(DeathsCOVID)$ and $ln(CasesCOVID)$, which are the log state-level COVID-related deaths and cases per capita. Standard errors (in parentheses) are clustered at individual and time levels. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

Table A.6: Estimated overreaction α

	(1)
θ	0.318**
	(0.142)
α	0.241
	(0.250)
$\alpha \times$ $Tenuretercile2$	0.089
	(0.063)
$\alpha \times$ $Tenuretercile3$	0.046
	(0.067)
$\alpha \times$ $College$	0.006
	(0.092)
$\alpha \times$ $Age : under40$	0.112***
	(0.038)
$\alpha \times$ $Highnumeracy$	-0.074*
	(0.040)
$\bar{g}$	0.030*
	(0.018)
Sample	Jun13–Feb20
Observations	768

Notes:

Table A.7: Estimated overreaction θ

	(1)
α	0.285*
	(0.172)
θ	0.270**
	(0.133)
$\theta \times$ $Tenuretercile2$	0.086
	(0.066)
$\theta \times$ $Tenuretercile3$	0.050
	(0.079)
$\theta \times$ $College$	0.013
	(0.115)
$\theta \times$ $Age : under40$	0.130***
	(0.049)
$\theta \times$ $Highnumeracy$	-0.075
	(0.055)
$\bar{g}$	0.029*
	(0.017)
Sample	Jun13–Feb20
Observations	768

Notes:

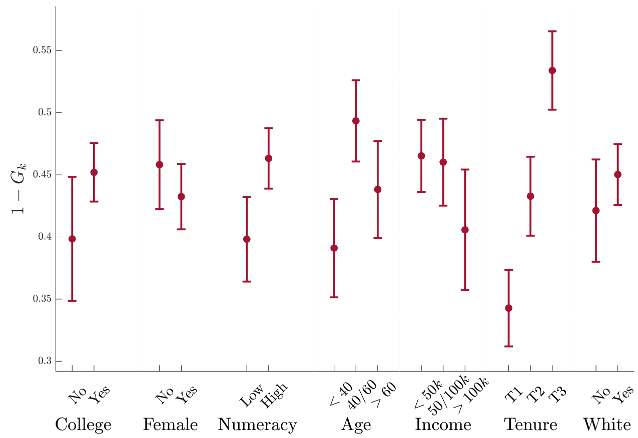
Table A.8: Belief stickiness and uncertainty

	(1) $1 - G_{j,t}$	(2) $1 - G_{j,t}$	(3) $1 - G_{j,t}$
$ln(\sigma_{j,t+h,t-1})$	-0.261***	-0.262***	-0.262***
	(0.045)	(0.045)	(0.045)
$ln(\widehat{\sigma}_{e,j,t})$	0.355***	0.354***	0.354***
	(0.028)	(0.028)	(0.028)
$\pi_{t,t-12}$	0.004		
	(0.009)		
$\pi_{t,t-1}$		-0.011	
		(0.040)	
$\pi_{t-1,t-2}$			-0.004
			(0.038)
Year-Month FEs	N	N	N
Group FE	Y	Y	Y
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Adjusted R-squared	0.63	0.63	0.63
Observations	768	768	768

Notes: Standard errors (in parentheses) are bootstrapped within and across group-month (j, t) level. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

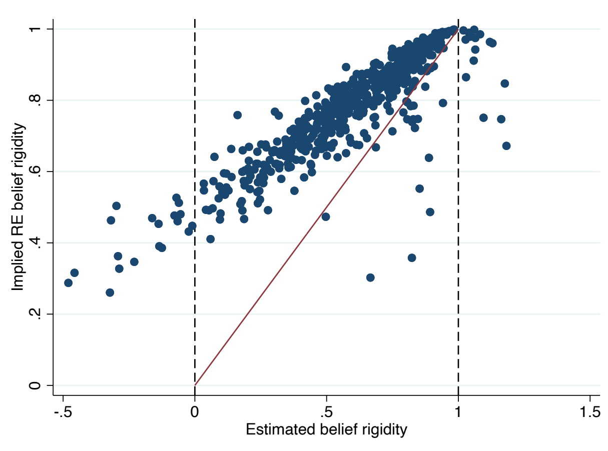
F Additional figures

Figure A.2: Belief updating across groups: $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$: only households revising their answer from previous month



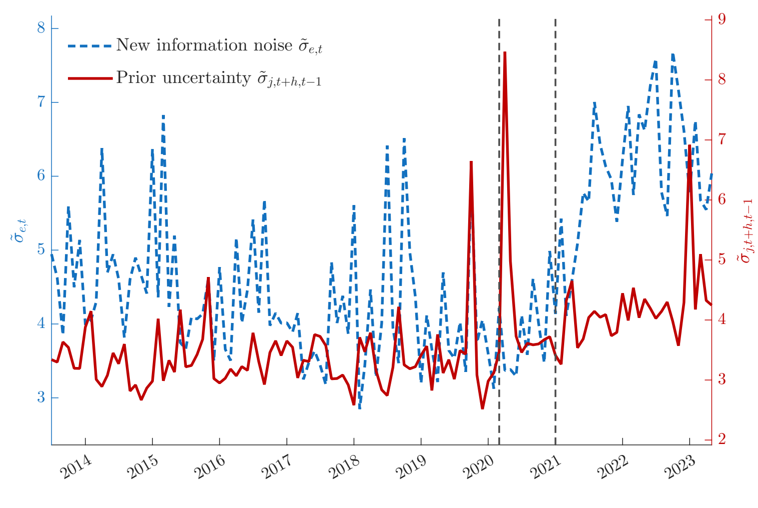
Notes: The figure reports the group-specific belief stickiness from (9), $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$, estimated including socioeconomic groups individually in the regression. This figure only considers households providing in t a different forecast than in the previous month. Sample: 2013M6-2023M5. Bars represent the 95% confidence interval.

Figure A.3: Belief stickiness and uncertainty: consumer with large tenure



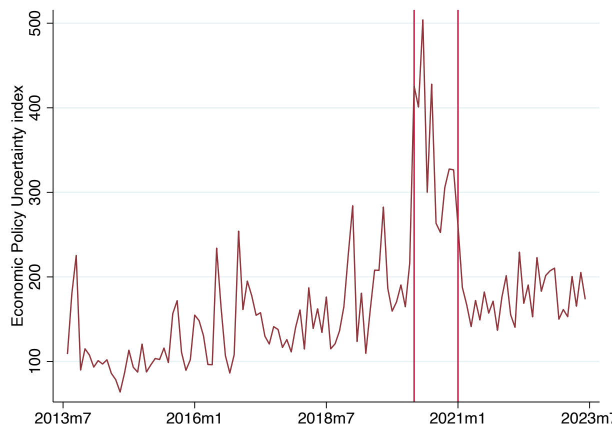
Notes: The figure plots on the x-axis the estimated belief stickiness $\hat{\beta}_{j,t}$ from regression (7) against the implied rational expectation belief stickiness from equation (10) on the y-axis. The sample excludes consumers in the first quartile of the tenure distribution.

Figure A.4: Decomposing uncertainty: prior and new information noise under Rational Expectation



Notes: The figure plots the estimates of new information noise (left y-axis) and prior uncertainty (right y-axis) from equation (16), with the estimates of estimates of belief stickiness $\widehat{1 - G_t}$, posterior uncertainty $\hat{\sigma}_{j,t+h,t}$ and $\alpha = 0, \theta = 0$.

Figure A.5: Economic Policy Uncertainty index



Notes: The figure plots the "news coverage component" from the Economic Policy Uncertainty Index, which is based on the share of articles in US online newspapers that mention economic policy uncertainty (Baker et al., 2016). This index is based on the share of articles in 10 large newspapers (USA Today, the Miami Herald, the Chicago Tribune, the Washington Post, the Los Angeles Times, the Boston Globe, the San Francisco Chronicle, the Dallas Morning News, the Houston Chronicle, and the WSJ) containing the terms 'uncertainty' or 'uncertain', the terms 'economic' or 'economy' and one or more of the following terms: 'congress', 'legislation', 'white house', 'regulation', 'federal reserve', or 'deficit'. For additional details, see Baker et al. (2016).

G Belief stickiness and uncertainty: alternative empirical strategy

We investigate the relationship between uncertainty and belief stickiness with a different empirical strategy compared to section 3. Instead of regressing prior and posterior uncertainty on the estimated group-month belief stickiness, we estimate belief stickiness and its relationship with uncertainty in the same regression by using an interaction term. That is, we regress

$$E_{i,t}[x_{t+h}] = \alpha + \beta_1 E_{i,t-1}[x_{t+h}] + \mathbf{X}_{i,t}\mathbf{B}_2 + E_{i,t-1}[x_{t+h}] \times \mathbf{X}_{i,t}\mathbf{B}_3 + \gamma_t + \beta_4(\gamma_t \times \mathbf{X}_{i,t}) + err_t^i \tag{A.2}$$

where $\mathbf{X}_{i,t}$ is a vector containing the measure of prior and a proxy for new information noise, namely posterior uncertainty or absolute forecast error. Table A.9 shows that the results are consistent with the ones in Table 1 and with Bayesian updating: belief stickiness is higher if new information is noisy and lower if the prior uncertainty is higher.

Table A.9: Belief stickiness and uncertainty

	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$	(4) $E_t^i[x_{t+h}]$
$E_{t-1}^i[x_{t+h}]$	0.523***	0.065**	0.467***	0.157***
	(0.024)	(0.032)	(0.023)	(0.023)
$E_{t-1}^i[x_{t+h}] \times ln(\sigma_{i,t+h,t-1})$	-0.122***	-0.128***		
	(0.016)	(0.012)		
$High(\sigma_{i,t+h,t-1})$=1 $\times$ $E_{t-1}^i[x_{t+h}]$			-0.188***	-0.126***
			(0.023)	(0.027)
$High(\sigma_{i,t+h,t-1})$=1 $\times$ $E_{t-1}^i[x_{t+h}]$				
$E_{t-1}^i[x_{t+h}] \times ln(\sigma_{i,t+h,t})$	0.106***			
	(0.015)			
$E_{t-1}^i[x_{t+h}] \times ln(abs(FE_{i,t}))$		0.250***		
		(0.013)		
$High(\sigma_{i,t+h,t})$=1 $\times$ $E_{t-1}^i[x_{t+h}]$			0.241***	
			(0.019)	
$High(abs(FE_{i,t}))$=1 $\times$ $E_{t-1}^i[x_{t+h}]$				0.488***
				(0.015)
Constant	2.532***	3.420***	2.344***	2.795***
	(0.069)	(0.089)	(0.057)	(0.052)
Year-Month FEs	Y	Y	Y	Y
Non-interacted variables	Y	Y	Y	Y
Year-Month $\times$ variables	Y	Y	Y	Y
Socioeconomic controls	Y	Y	Y	Y
Sample	Jun13-Feb20	Jun13-Feb20	Jun13-Feb20	Jun13-Feb20
Adjusted R-squared	0.39	0.50	0.37	0.39
Observations	55706	62786	65192	62786

Notes: This table reports the estimated coefficient from regression (A.2). We control for time fixed effect and their interaction with the variables. Standard errors (in parentheses) are clustered at the group-month level. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

H Aggregate belief stickiness

We compute belief stickiness by running the following panel regression in a rolling window of three months

$$E_t^i[x_{t+h}] = \alpha + \beta E_{t-1}^i[x_{t+h}] + \gamma_t + X_{i,t} + err_t^i \tag{A.3}$$

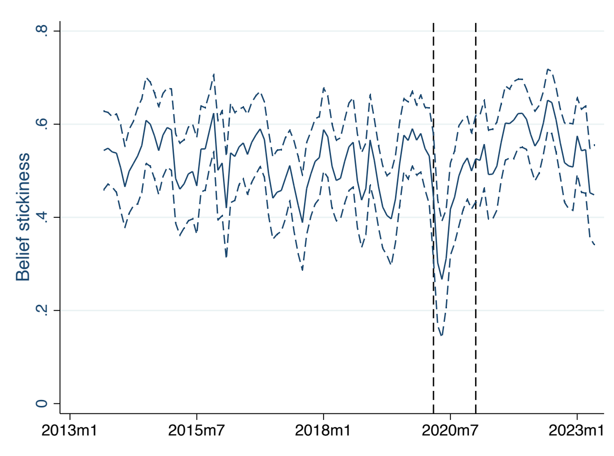
where $X_{i,t}$ contains sociodemographic controls, such as tenure (i.e. number of months in the survey), whether holding a college degree, age, income over 100k or below 50k, high numeracy, gender, and race. Figure A.6 plots the estimated belief stickiness $\hat{\beta}$ for each final month of the 3-month rolling window. Figure A.7 reports the same regression, but considering only the intensive margin of belief adjustment, meaning excluding observations where the posterior (forecast in t) equals the prior (forecast in $t-1$), with the same result.

However, in the presence of heterogeneity in belief updating across households, as captured in our model (4), estimating aggregate belief stickiness through (A.3) is biased. To see this, we demean posterior expectations in (4) by the aggregate (rather than group-specific) and get:

$$\begin{aligned} &E_{i,j,t}[x_{t+h}] - \int^i E_{i,j,t}[x_{t+h}]di = \\ &(1-G_{j,t})E_{i,j,t-1}[x_{t+h}] - \int^i (1-G_{j,t})E_{i,j,t-1}[x_{t+h}]di + (G_{j,t} - \int^i G_{j,t}di)x_{t+h} + G_{j,t}e_{i,j,t} \end{aligned} \tag{A.4}$$

which implies that $\hat{\beta}$ in (A.3) does not map directly to the average belief stickiness $\overline{1-G} = \int^i (1-G_j)di$. For this reason, we rely on the group-level regression in equation (7). Nonetheless, the dynamics around the COVID-19 outbreak look similar in both approaches, reinforcing the robustness of our baseline results.

Figure A.6: Belief stickiness pre- and post-pandemic



Notes: The figure plots the estimate of A.3 excluding observations where the posterior equals the prior, meaning excluding revisions equal to zero. The dashed blue lines represent the 95% confidence interval. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

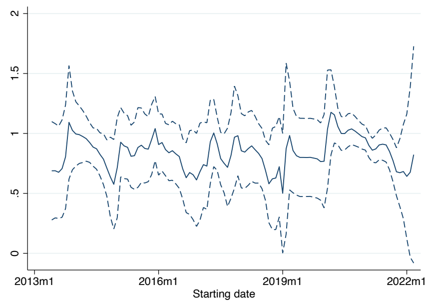
Figure A.7: Belief stickiness pre- and post-pandemic (non-zero revisions)

Notes: The figure plots the estimate of A.3, while the dashed blue lines represent the 95% confidence interval. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

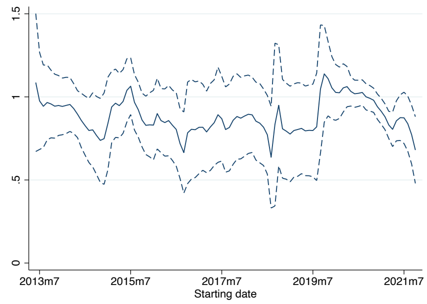
I Estimation of inflation autoregressive structure

Figure A.8: Rolling window autoregressive estimation of monthly inflation

(a) 15-month window



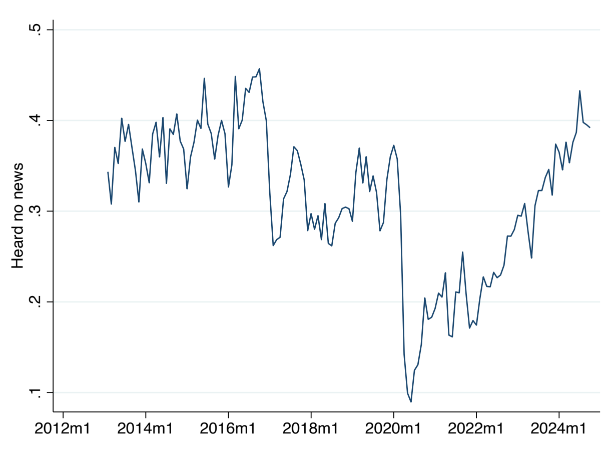
(b) 20-month window



Notes: This figure show the estimates of an AR(1) autoregressive process for monthly inflation in a rolling window.

J Michigan Survey of Consumers

Figure A.9: Share of respondents who heard no news about Business Conditions



We consider a question in the Michigan Survey of Consumers: "During the last few months, have you heard of any favorable or unfavorable changes in business conditions?". In each month, we compute the share of respondents that answer "No". Figure A.9 reports this share, which declines at the onset of the COVID-19 pandemic, only to increase back in the following periods. While we can distinguish whether this is driven by demand (consumer) or supply (media) factors, it is nevertheless consistent with the evidence that consumers look for new information during the COVID period and less afterwards.

K Lockdowns and Belief Formation

In this appendix, we investigate the role played by lockdown policies in driving the decline in the updating gain documented during the pandemic. The restrictions on movement and activity implemented to stop the spread of the virus may have shaped both households' belief updating and their subjective uncertainty.

K.1 Measuring Lockdown Stringency

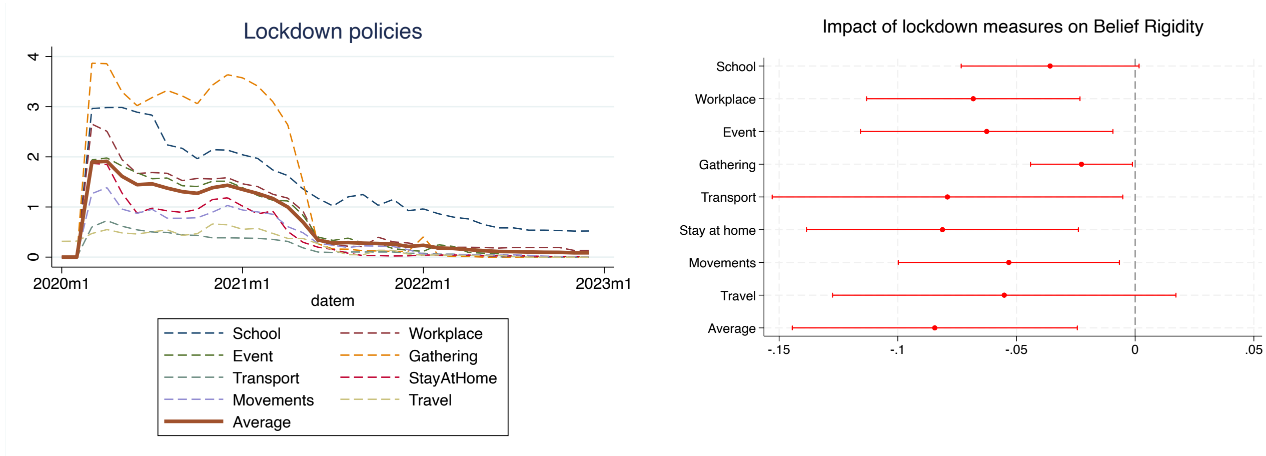
We measure the US state-level stringency of lockdown policies from the Oxford COVID-19 Government Response Tracker (OxCGRT) database. The database covers the period between January 2020 and December 2022 and contains information about closure and containment restrictions, which are recorded as ordinal categorical scales measuring the intensity or severity of the policy. Details about the collection process for a variety of countries are in Hale et al. (2020), while Hallas et al. (2021) provides an overview of the policy implemented at the US state level. We consider the following indicators: *school closing*, *workplace closing*, *cancel public events*, *restrictions on gathering size*, *close public transport*, *stay at home requirements*, and *restrictions on internal movements*. As the severity of these policies differs between vaccinated and non-vaccinated individuals, we consider the state average weighted by the number of vaccinated and non-vaccinated individuals. Finally, we compute a summary measure of the severity of lockdown measures, *lockdown*, equal to the simple average of these indicators.[28] Figure A.10 reports the time series of the country-level average of each indicator. Moreover, to measure the local impact of the pandemic we use the US state-level monthly level of COVID deaths and cases per capita. Table A.4 reports the summary statistics.

K.2 Impact of Lockdowns on the Updating Gain

To estimate the effect of lockdown measures on the updating gain, we interact prior inflation forecasts with the lockdown index and COVID-19 case and death measures. This approach allows us to isolate the impact of lockdown policies from the effects of COVID-19's direct health impact. We run the following regression:

[28]This measure is similar to the *stringency index* in Hale et al. (2020), as they also consider a simple average of each indicator. However, differently from them, we exclude from this average the indicators on *restrictions on international travel*, as not related to state-level measures, and *public information campaign*, as not related to lockdown measures.

Figure A.10: Lockdown policies and the updating gain



Notes: The left figure represents the average state-level lockdown policy intensity for different social activities, weighted by state population. The data source for lockdowns is the Oxford Covid-19 Government Response Tracker (OxCGRT). The right plot shows the impact of lockdown measures on the updating gain, β_2 in (A.5). Sample period: 2020M3-2022M12.

$$
\begin{aligned}
E_t^i[x_{t+h}] =& \alpha + \beta_1 E_{t-1}^i[x_{t+h}] + \beta_2 E_{t-1}^i[x_{t+h}] \times Lockdown_{k,t} + \beta_3 Lockdown_{k,t} + \\
&+ E_{t-1}^i[x_{t+h}] \times CovidSeverity_{k,t}\Pi + CovidSeverity_{k,t}\Gamma + \\
&+ \lambda X_{i,t} + [CovidSeverity_{k,t} \quad Lockdown_{k,t}] \times \gamma_t + err_t^i,
\end{aligned}
\tag{A.5}
$$

where $Lockdown_{k,t}$ contains the lockdown indexes, while $CovidSeverity_{k,t}$ contains the COVID cases and deaths in state k at date t. The key coefficient of interest, β_2, captures how lockdown policies influence the updating gain. We run the regression in the post-pandemic sample, from March 2020.

Figure A.10 reports the estimates of β_2, while Table A.5 reports the detailed results. All lockdown indicators have a robust negative effect on $1 - G_{j,t}$, meaning they increase the updating gain. While all individual indicators are significant, including all of them together may create collinearity issues. We therefore use the average of the indices as a summary measure. The impact remains negative and robust. This result suggests that lockdown policies lowered the cost of collecting information for consumers, leading them to update their beliefs more.

Table A.10 presents additional evidence. The first column replicates the last column of Table A.5, using the average index *Lockdown* to summarize the stringency of state-level lockdown policies. The impact on the updating gain is still robust. In the next three columns, we compare the effect of lockdown policies with measures of state-level economic policy uncertainty from Baker et al. (2022). The indices are constructed from articles in local

Table A.10: Updating gain and lockdown measures

	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$	(4) $E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.600***	0.604***	0.606***	0.603***
	(0.113)	(0.116)	(0.119)	(0.118)
$E_{i,t-1}[x_{t+h}] \times Lockdown$	-0.082***	-0.083***	-0.083***	-0.082***
	(0.030)	(0.030)	(0.030)	(0.030)
$E_{i,t-1}[x_{t+h}] \times ln(DeathsCOVID)$	-0.017	-0.016	-0.016	-0.016
	(0.016)	(0.016)	(0.016)	(0.016)
$E_{i,t-1}[x_{t+h}] \times ln(CasesCOVID)$	0.032*	0.032*	0.032*	0.032*
	(0.016)	(0.016)	(0.016)	(0.016)
$E_{i,t-1}[x_{t+h}] \times \Delta ln(EPUState)$		0.010		
		(0.014)		
$E_{i,t-1}[x_{t+h}] \times \Delta ln(EPUNational)$			0.005	
			(0.016)	
$E_{i,t-1}[x_{t+h}] \times \Delta ln(EPUComposite)$				0.007
				(0.021)
Constant	2.451***	2.451***	2.453***	2.452***
	(0.085)	(0.085)	(0.085)	(0.085)
Year-Month FEs	Y	Y	Y	Y
Socio-democraphic FEs	Y	Y	Y	Y
Non-interacted variables	Y	Y	Y	Y
Sample	Mar20-May23	Mar20-May23	Mar20-May23	Mar20-May23
Adjusted R-squared	0.35	0.35	0.35	0.35
Observations	25398	25397	25381	25398

Notes: For denotes the 1-year ahead forecast of inflation expectations starting 24 months into the future from the NY Fed Survey of Consumer Expectations (SCE). $Prior$ is the point forecast about the 3-year horizon provided in the previous month. $DeathsCOVID$ and $CasesCOVID$ are respectively the state-level COVID-related deaths and cases per capita. The $EPUstate$, $National$, and $Composite$ are the state-level economic policy uncertainty indicators from Baker et al. (2022). We control for year-month fixed effects and their interaction with the other variables. Standard errors (in parentheses) are clustered at individual and time levels. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

newspapers containing terms such as 'economic' and 'uncertainty', and are divided according to the topic of the economic policy considered: national-level, state-level, and a composite of the two.[29] Even controlling for state-level uncertainty, the estimated impact of lockdown policies on the updating gain is significant.[30]

K.3 Impact of Lockdowns on Uncertainty

The documented negative impact of lockdown restrictions on $1 - G_{j,t}$ is consistent with a decline in the cost of collecting information. From the lens of our general framework in Section 1.1, a lower marginal cost of information collection corresponds to a decrease in new information noise $\sigma_{e,t}^2$ (Maćkowiak et al., 2023; Pomatto et al., 2023). Higher new information accuracy would then not only increase the updating gain, but also reduce posterior

[29] We take the percentage change in the measure to isolate the surprise component. The results are robust to using simple differences and levels.

[30] Table A.14 reports the results for the one-year inflation forecasts, with similar results.

uncertainty.[31] To test this empirically, we estimate the following regression at the US state level:

$$\sigma_{k,t+h,t} = \alpha + \beta Lockdown_{k,t} + CovidImpact'_{k,t}\Gamma + \delta ln(EPU)_{k,t} + \gamma_j + err_{k,t} \tag{A.6}$$

where $\sigma_{k,t+h,t} = \int_{i \in k} \sigma_{i,t+h,t} di$ denotes the average posterior uncertainty for consumers in state k at time t. The key variable of interest is $Lockdown_{k,t}$, the average index of lockdown intensity. Table A.11 reports the estimated coefficients.

Results show a robust and negative effect of lockdown policies on posterior belief uncertainty, consistent with the idea that lower information costs allow for better information gathering. Economic policy uncertainty ($EPU_{k,t}$) increases posterior uncertainty, as expected.

Table A.11: Uncertainty and lockdown measures

	(1) $\sigma_{k,t+h,t}$	(2) $\sigma_{k,t+h,t}$	(3) $\sigma_{k,t+h,t}$
Lockdown	-0.379***	-0.362***	-0.234***
	(0.084)	(0.084)	(0.063)
$\sigma_{k,t+h-1,t-1}$			0.464***
			(0.034)
$ln(DeathsCOVID)$			0.017
			(0.036)
$ln(CasesCOVID)$			-0.028
			(0.037)
$ln(EPUNational)$			0.068**
			(0.032)
Constant	3.569***	3.557***	1.621***
	(0.103)	(0.052)	(0.313)
State FEs	N	Y	Y
Sample	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22
Adjusted R-squared	0.05	0.28	0.43
Observations	1717	1717	1694

Notes: The table reports the results from regression (A.6). $EPUComposite$ is the state-level economic policy uncertainty indicator from Baker et al. (2022). We control for state FEs. Covid controls are logged $DeathsCOVID$ and $CasesCOVID$, which are the state-level COVID-related deaths and cases per capita. Standard errors (in parentheses) are clustered at State and time levels. $\star$, $\star\star$, and $\star\star\star$ are 10%, 5%, and 1% confidence levels.

These findings show that lockdown policies significantly increased the updating gain while

[31] An alternative possibility is that lower information costs led to higher, instead of lower, belief uncertainty. This could be the case, for example, if consumers could learn about signals' accuracy only by acquiring more signals. In this case, a lower information cost would allow consumers to acquire more signals and learn about the increase in the signal's noise, which could explain both the higher updating gain and the higher belief uncertainty.

also lowering posterior uncertainty, indicating that consumers perceived the information they gathered as more precise. This is consistent with theoretical predictions about the effects of lower information costs on belief formation.[32]

At the aggregate level, however, reduced information costs can account for the increase in the updating gain during the pandemic, but cannot explain the simultaneous rise in uncertainty. As discussed in Section 4, only an increase in fundamental volatility can reconcile both empirical patterns.

L Belief uncertainty and disagreement

In this section we clarify the difference between belief uncertainty, namely the variance of posterior beliefs, and disagreement, meaning the cross-sectional dispersion in posterior mean. We show that new information noise unambiguously increases the former, but not the latter.

Simple model Consider a simple version of the belief-updating setting considered in section **??** where the variable forecasted is i.i.d. Suppose agents form beliefs about stochastic variable $x \sim N(\mu_x, \sigma_x^2)$ where μ_x is the prior mean and σ_x^2 is the prior variance. Agents can not observe x directly, but receive a private noisy signal about it, similarly to equation (2)

$$s^i = x + e^i \tag{A.7}$$

where the signal noise $e^i = \eta^i + \omega$ contains (i) an idiosyncratic component η^i normally distributed mean-zero noise with variance σ_η^2 and i.i.d. across time and households, i.e. $\int^i e_t^i di = 0$, and (ii) a common component ω normally distributed mean-zero noise with variance σ_ω^2 which is i.i.d. only across time, but not across agents. Let $\sigma_e^2 \equiv \sigma_\eta^2 + \sigma_\omega^2$ define the overall variance of the signal noise.

The Bayesian posterior beliefs is $x|s^i \sim N(E^i[x|s^i], Var^i[x|s^i])$. The posterior mean equals

$$E^i[x|s^i] = (1 - G)\mu_x + Gs^i \tag{A.8}$$

where the Bayesian weight on new information is $G = \frac{\sigma_x^2}{\sigma_e^2+\sigma_x^2} = \frac{\sigma_x^2}{\sigma_\eta^2+\sigma_\omega^2+\sigma_x^2}$.

[32] Our findings are also consistent with a negative effect of lockdown policies on state-level fundamental uncertainty, which households may have extrapolated to aggregate conditions. Such a mechanism would likewise reduce perceived uncertainty and increase the updating gain.

Belief uncertainty The Bayesian posterior variance, or uncertainty, then equals

$$Var^i[x|s^i] = \sigma_e^2 G = \frac{\sigma_x^2 \sigma_e^2}{\sigma_e^2 + \sigma_x^2} \tag{A.9}$$

Therefore

$$\frac{\partial Var^i[x|s^i]}{\partial \sigma_e^2} = \left(\frac{\sigma_x^2}{\sigma_e^2 + \sigma_x^2} \right)^2 > 0 \tag{A.10}$$

Posterior uncertainty increases in the new information noise, no matter whether the increase is due to new private information noise σ_η^2 or new public information noise σ_ω^2. In section **??**, we do not need to take a stand about what drives the increase in new information noise to derive our main implications.

Belief disagreement Let's consider now disagreement, meaning cross-sectional dispersion of posterior mean across agents.

$$Disp(E^i[x|s^i]) = G^2 \sigma_\eta^2 \tag{A.11}$$

As we consider the second moment of the cross-sectional distribution, the common error and realization across forecasters drop out.

First, consider an increase in public information noise σ_ω^2. Intuitively, there is no direct effect on disagreement, as new information received by agents becomes equally more uncertain. If all information were public, then there would be no effect at all on disagreement. However, since the new signal also contains private information, an increase in public noise has an indirect effect on belief dispersion: as the new signal is overall noisier, agents allocate less weight G to it, and more to the common prior, leading to a decrease in disagreement:

$$\frac{\partial Disp(E^i[x|s^i])}{\partial \sigma_\omega^2} = \sigma_\eta^2 \frac{\partial G^2}{\partial \sigma_\omega^2} = -2 \frac{\sigma_x^2 \sigma_\eta^2}{(\sigma_\eta^2 + \sigma_\omega^2 + \sigma_x^2)^2} < 0 \tag{A.12}$$

Now consider an increase in private information noise σ_η^2. In this case, there are two effects. First, a direct effect: larger volatility of idiosyncratic shocks makes new information more dispersed across agents. This is represented by the first term on the right-hand side of equation (A.13). Second, an indirect effect: as new information is overall noisier, agents allocate less weight to G to it, and more to the common prior. This is represented by the second term on the right-hand side of equation (A.13)

$$\frac{\partial Disp(E^i[x|s^i])}{\partial \sigma_\eta^2} = G^2 + \sigma_\eta^2 \frac{\partial G^2}{\partial \sigma_\eta^2} \tag{A.13}$$

Therefore,

$$\frac{\partial Disp(E^i[x|s^i])}{\partial \sigma_\eta^2} > 0 \Longleftrightarrow \sigma_\eta^2 < \frac{\sigma_x^2}{2} \tag{A.14}$$

The first effect prevails for low values of σ_η^2, while the second prevails for higher values of σ_η^2. In other words, the effect of new private information noise on belief dispersion is non-monotonic.

To sum up, while an increase in new information noise unambiguously increases posterior uncertainty, the effect on belief disagreement is nuanced and can go in either direction.

M Belief stickiness and forecast errors

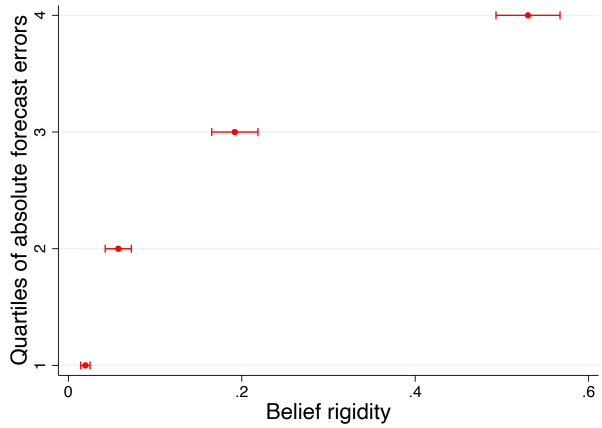
We investigate whether higher belief stickiness is associated with higher forecast errors. We define forecast error as the difference between the realization of 12-month inflation starting 24 months from the survey data and the individual forecast. We run the following regression

$$E_t^i[x_{t+h}] = \alpha + \beta_1 E_{t-1}^i[x_{t+h}] + \mathbf{FE}_{i,t}\mathbf{B}_2 + E_{t-1}^i[x_{t+h}] \times \mathbf{FE}_{i,t}\mathbf{B}_3 + \gamma_t + \beta_4(\gamma_t \times \mathbf{FE}_{i,t}) + err_t^i \tag{A.15}$$

where $FE_{i,t}$ contains the dummy indicators with value 1 if the absolute forecast error of individual i at time t is in the first, second, third, or fourth quartile of the unconditional absolute forecast error distribution. The coefficient $\mathbf{B}_3$ identifies belief stickiness at different forecast error quartiles.

Figure A.11 reports the results. We find that the relation between belief stickiness and forecast errors is positive: higher belief stickiness is associated with higher forecast errors. Table A.12 reports alternative specifications. The results are robust to using absolute forecast errors and limiting the sample to pre-Covid period.

Figure A.11: Forecast errors



Notes: The figure shows the relationship between forecast errors and belief stickiness, $\mathbf{B}_3$ from regression A.15 as reported in column (3) of table A.12. Bars represent the 95% confidence interval. Sample period: 2013M6-2019M11.

Table A.12: Forecast errors and belief stickiness

	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.566***	0.180***	0.020***
	(0.012)	(0.013)	(0.003)
$E_{i,t-1}[x_{t+h}] \times abs(FE)$		0.008***	
		(0.001)	
Quartile $abs(FE)$=2 $\times$ $E_{i,t-1}[x_{t+h}]$			0.058***
			(0.008)
Quartile $abs(FE)$=3 $\times$ $E_{i,t-1}[x_{t+h}]$			0.192***
			(0.014)
Quartile $abs(FE)$=4 $\times$ $E_{i,t-1}[x_{t+h}]$			0.530***
			(0.019)
Constant	1.981***	3.530***	3.243***
	(0.057)	(0.087)	(0.044)
Year-Month FEs	Y	Y	Y
Year-Month FEs $\times$ FE	N	Y	Y
Non-interacted FE	Y	Y	Y
Sample	Jun13-Dec19	Jun13-Dec19	Jun13-Dec19
Adjusted R-squared	0.32	0.59	0.43
Observations	63509	63509	63509

Notes: $abs(FE)$ indicates the absolute forecast error of the individual, defined as the absolute difference between the 12-month inflation starting 24 months into the future and $Forecast$. *Quartile* $abs(FE)$ equals 1 if $abs(FE)$ lies in the specific quartile of the unconditional distribution of $abs(FE)$. The estimation period is 2013M6-2023M5. Standard errors (in parentheses) are clustered at the individual and time levels. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

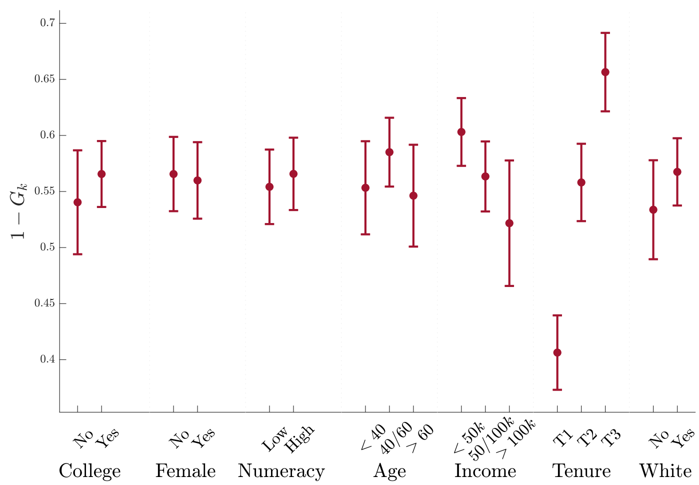
Table A.13: Belief stickiness and uncertainty: 1-year horizon

	(1) $1 - G_{j,t}$	(2) $1 - G_{j,t}$	(3) $1 - G_{j,t}$	(4) $1 - G_{j,t}$
$ln(\sigma_{j,t+h,t-1})$	-0.207***		-0.393***	-0.250***
	(0.034)		(0.029)	(0.046)
$ln(\widehat{\sigma}_{e,j,t})$		0.282***	0.335***	0.325***
		(0.021)	(0.027)	(0.028)
Year-Month FEs	Y	Y	Y	Y
Group FE	N	N	N	Y
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Adjusted R-squared	0.13	0.46	0.62	0.64
Observations	737	737	737	737

Notes: This table reports the estimated coefficient from regression (13). Standard errors (in parentheses) are computed with a two-level bootstrap. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

N Shorter forecast horizon

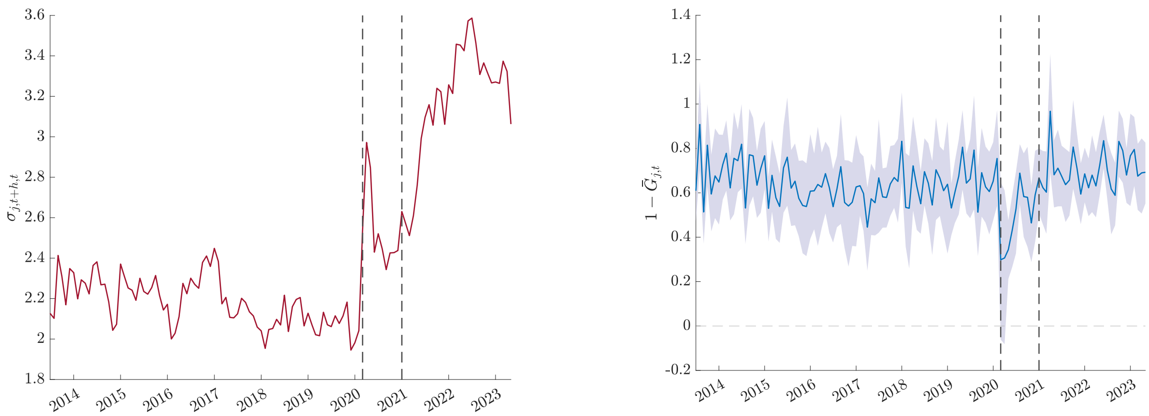
Figure A.12: Belief updating across groups: $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$: 1-year horizon



Notes: The figure reports the group-specific belief stickiness from (9), $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$, estimated including socioeconomic groups individually in the regression. Expectations are about the 12-month ahead (1-year) inflation. Sample: 2013M6-2023M5. Bars represent the 95% confidence interval.

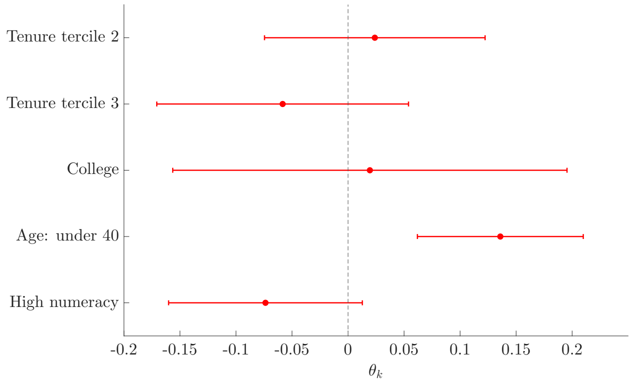
Figure A.13: Belief stickiness pre- and post-pandemic: 1-year horizon

(a) Average uncertainty (b) Average belief stickiness

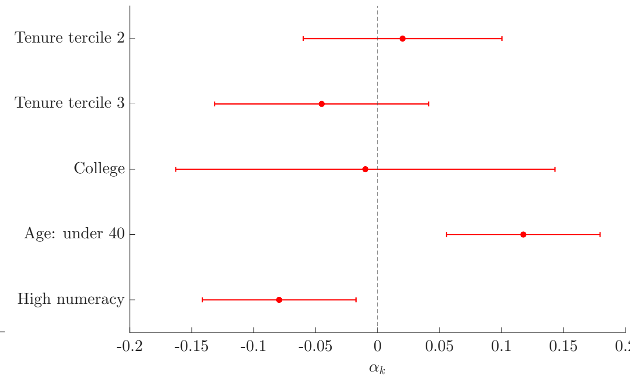


Notes: The left panel plots the average posterior belief uncertainty (standard deviation) across all consumers. The right panel plots the average belief stickiness across groups from regression 7, while the dashed blue lines represent the average 90% confidence interval. Standard errors are computed with a two-level bootstrap, within and across *(j,t)* groups. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

Figure A.14: Heterogeneity in overreaction bias: 1-year horizon



(a) Heterogeneity in θ



(b) Heterogeneity in α

Notes: The left panel plots the estimated interactions with socioeconomic dummies and θ in 15. The right panel plots the estimated interactions with socioeconomic dummies and α in 15. Confidence intervals are at the 90% level and standard errors are computed with a two-level bootstrap, within and across *(j,t)* groups. Sample Jun13-Feb20.

Table A.14: Belief stickiness and lockdown measures: : 1-year horizon

	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$	(4) $E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.788***	0.766***	0.789***	0.777***
	(0.213)	(0.190)	(0.206)	(0.194)
$E_{i,t-1}[x_{t+h}] \times Lockdown$	-0.109***	-0.107***	-0.109***	-0.108***
	(0.027)	(0.026)	(0.027)	(0.026)
$E_{i,t-1}[x_{t+h}] \times ln(DeathsCOVID)$	0.008	0.007	0.008	0.008
	(0.021)	(0.019)	(0.020)	(0.019)
$E_{i,t-1}[x_{t+h}] \times ln(CasesCOVID)$	0.017	0.015	0.017	0.016
	(0.016)	(0.016)	(0.016)	(0.016)
$E_{i,t-1}[x_{t+h}] \times \Delta ln(EPUState)$		-0.020		
		(0.025)		
$E_{i,t-1}[x_{t+h}] \times \Delta ln(EPUNational)$			0.004	
			(0.020)	
$E_{i,t-1}[x_{t+h}] \times \Delta ln(EPUComposite)$				-0.012
				(0.030)
Constant	3.099***	3.103***	3.100***	3.102***
	(0.121)	(0.121)	(0.122)	(0.122)
Year-Month FEs	Y	Y	Y	Y
Socio-demographic FEs	Y	Y	Y	Y
Non-interacted variables	Y	Y	Y	Y
Sample	Mar20-May23	Mar20-May23	Mar20-May23	Mar20-May23
Adjusted R-squared	0.40	0.40	0.40	0.40
Observations	25263	25261	25248	25263

Notes: *DeathsCOVID* and *CasesCOVID* are respectively the state-level COVID-related deaths and cases per capita. The *EPUstate*, *National*, and *Composite* are the state-level economic policy uncertainty indicators from Baker et al. (2022). We control for year-month fixed effects and their interaction with the other variables. Standard errors (in parentheses) are clustered at individual and time levels. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

Table A.15: Belief stickiness and lockdown measures: 1-year horizon

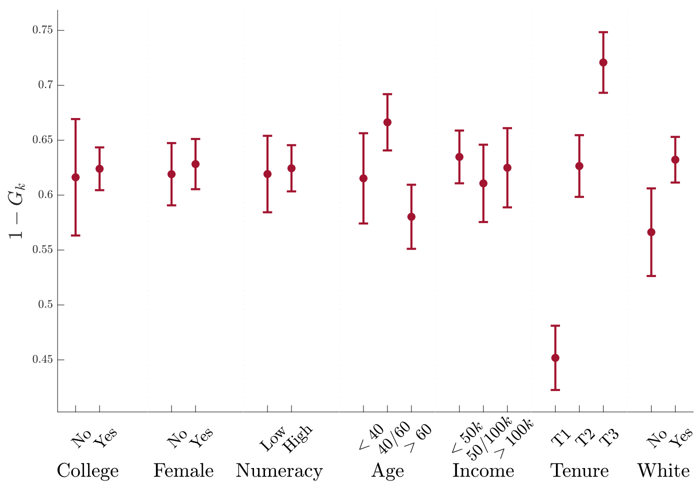
	(1)	(2)	(3)
	$\sigma_{k,t+h,t}$	$\sigma_{k,t+h,t}$	$\sigma_{k,t+h,t}$
Lockdown	-0.468***	-0.459***	-0.320***
	(0.085)	(0.083)	(0.064)
$\sigma_{k,t+h-1,t-1}$			0.415***
			(0.044)
$ln(DeathsCOVID)$			0.011
			(0.043)
$ln(CasesCOVID)$			-0.046
			(0.033)
$ln(EPUNational)$			0.070*
			(0.037)
Constant	3.754***	3.748***	1.753***
	(0.099)	(0.053)	(0.362)
State FEs	N	Y	Y
Sample	Mar20-Dec22	Mar20-Dec22	Mar20-Dec22
Adjusted R-squared	0.07	0.29	0.42
Observations	1718	1718	1696

Notes: The fable reports the results from regression A.6. $EPUComposite$ is the state-level economic policy uncertainty indicator from Baker et al. (2022). We control for state FEs. Covid controls are logged $DeathsCOVID$ and $CasesCOVID$, which are the state-level COVID-related deaths and cases per capita. Standard errors (in parentheses) are clustered at State and time levels. $\star$, $\star\star$, and $\star\star\star$ are 10%, 5%, and 1% confidence levels.

O Alternative measure of expected inflation: mean of subjective density forecasts

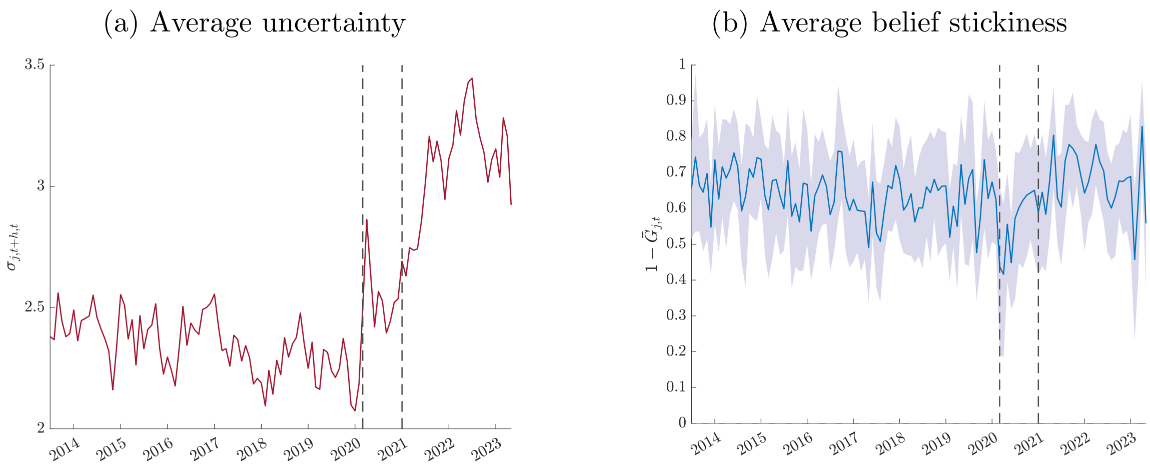
In the main text we use the point forecasts to measure expected mean inflation, i.e. the first moment of the subjective belief distribution, and the density forecasts uniquely to measure uncertainty, i.e. the second moment. Here, we instead measure the expected mean using the implied mean of the individual density forecasts.

Figure A.15: Belief updating across groups: $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$: expectations from mean of density forecasts



Notes: The figure reports the group-specific belief stickiness from (9), $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$, estimated including socioeconomic groups individually in the regression. Expectations are not measured as the mean of the density forecast, as described in section. Sample: 2013M6-2023M5. Bars represent the 95% confidence interval.

Figure A.16: Belief stickiness pre- and post-pandemic: expectations from mean of density forecasts



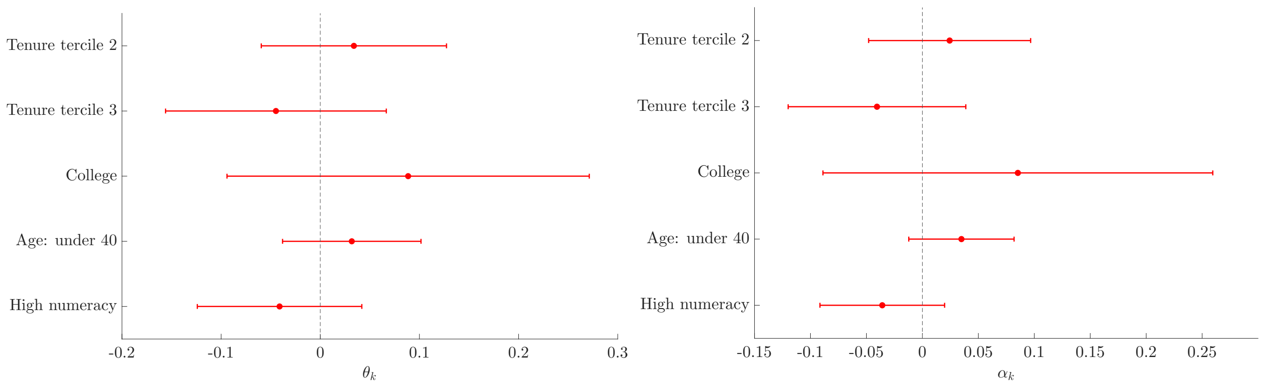
Notes: The left panel plots the average posterior belief uncertainty (standard deviation) across all consumers. The right panel plots the average belief stickiness across groups from regression 7, while the dashed blue lines represent the average 90% confidence interval. Standard errors are computed with a two-level bootstrap, within and across *(j,t)* groups. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

Table A.16: Belief stickiness and uncertainty: expectations from mean of density forecasts

	(1) $1 - G_{j,t}$	(2) $1 - G_{j,t}$	(3) $1 - G_{j,t}$	(4) $1 - G_{j,t}$
$ln(\sigma_{j,t+h,t-1})$	-0.207***		-0.393***	-0.250***
	(0.034)		(0.029)	(0.046)
$ln(\widehat{\sigma}_{e,j,t})$		0.282***	0.335***	0.325***
		(0.021)	(0.027)	(0.028)
Year-Month FEs	Y	Y	Y	Y
Group FE	N	N	N	Y
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Adjusted R-squared	0.13	0.46	0.62	0.64
Observations	737	737	737	737

Notes: This table reports the estimated coefficient from regression (13). Standard errors (in parentheses) are computed with a two-level bootstrap. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

Figure A.17: Heterogeneity in overreaction bias: expectations from mean of density forecasts



(a) Heterogeneity in θ (b) Heterogeneity in α

Notes: The left panel plots the estimated interactions with socioeconomic dummies and θ in 15. The right panel plots the estimated interactions with socioeconomic dummies and α in 15. Confidence intervals are at the 90% level and standard errors are computed with a two-level bootstrap, within and across *(j,t)* groups. Sample Jun13-Feb20.

Table A.17: Belief stickiness and lockdown measures: expectations from mean of density forecasts

	(1) $E_t^i[x_{t+h}]$	(2) $E_t^i[x_{t+h}]$	(3) $E_t^i[x_{t+h}]$	(4) $E_t^i[x_{t+h}]$
$E_{i,t-1}[x_{t+h}]$	0.938***	0.970***	0.955***	0.974***
	(0.130)	(0.128)	(0.135)	(0.138)
$E_{i,t-1}[x_{t+h}] \times$ *Lockdown*	-0.067**	-0.068**	-0.068**	-0.070**
	(0.028)	(0.029)	(0.028)	(0.028)
$E_{i,t-1}[x_{t+h}] \times$ *ln(DeathsCOVID)*	0.019	0.017	0.019	0.021
	(0.019)	(0.018)	(0.019)	(0.020)
$E_{i,t-1}[x_{t+h}] \times$ *ln(CasesCOVID)*	0.020	0.028*	0.021	0.022
	(0.016)	(0.015)	(0.016)	(0.017)
$E_{i,t-1}[x_{t+h}] \times \Delta$*ln(EPUState)*		0.029		
		(0.021)		
$E_{i,t-1}[x_{t+h}] \times \Delta$*ln(EPUNational)*			0.019	
			(0.017)	
$E_{i,t-1}[x_{t+h}] \times \Delta$*ln(EPUComposite)*				0.051**
				(0.021)
Constant	1.544***	1.536***	1.542***	1.542***
	(0.070)	(0.065)	(0.069)	(0.069)
Year-Month FEs	Y	Y	Y	Y
Socio-democraphic FEs	Y	Y	Y	Y
Non-interacted variables	Y	Y	Y	Y
Sample	Mar20-May23	Mar20-May23	Mar20-May23	Mar20-May23
Adjusted R-squared	0.37	0.37	0.37	0.37
Observations	26007	26004	25990	26007

Notes: *DeathsCOVID* and *CasesCOVID* are respectively the state-level COVID-related deaths and cases per capita. The *EPUstate*, *National*, and *Composite* are the state-level economic policy uncertainty indicators from Baker et al. (2022). We control for year-month fixed effects and their interaction with the other variables. Standard errors (in parentheses) are clustered at the individual and time levels. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

Table A.18: Belief stickiness and uncertainty: alternative calibration for prior expectations

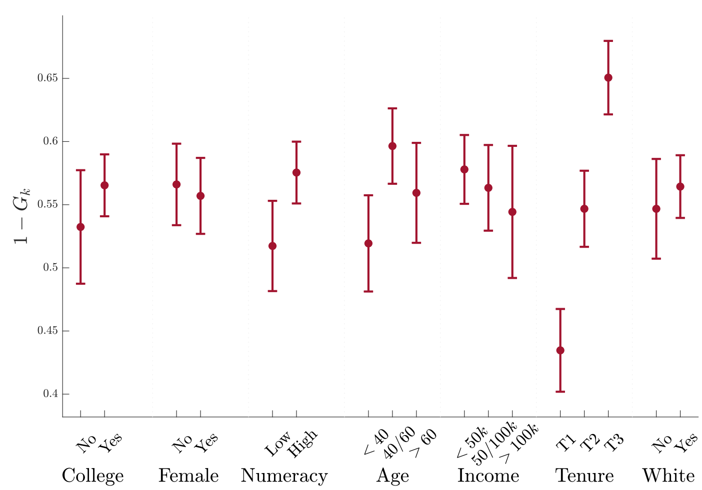
	(1) $1-G_{j,t}$	(2) $1-G_{j,t}$	(3) $1-G_{j,t}$	(4) $1-G_{j,t}$
$ln(\sigma_{j,t+h,t-1})$	-0.207***		-0.393***	-0.250***
	(0.034)		(0.029)	(0.046)
$ln(\widehat{\sigma}_{e,j,t})$		0.282***	0.335***	0.325***
		(0.021)	(0.027)	(0.028)
Year-Month FEs	Y	Y	Y	Y
Group FE	N	N	N	Y
Sample	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20	Jun13–Feb20
Adjusted R-squared	0.13	0.46	0.62	0.64
Observations	737	737	737	737

Notes: This table reports the estimated coefficient from regression (13). Standard errors (in parentheses) are computed with a two-level bootstrap. $*$ represents $p < 0.10$, $**$ represents $p < 0.05$, and $***$ represents $p < 0.01$.

P Alternative calibration for prior forecast

In the main text, we construct prior mean 4 and variance 11 using the estimated parameters $\rho = 0.946$, $\hat{\mu}_x = 2.3$ and $\sigma_u = 0.38$, reported in the first column of Table A.1. Here, we use an alternative calibration for this parameters, as reported in the second column of Table A.1: $\rho = 0.975$, $\mu_x = 2.82$ and $\sigma_u = 0.4$, with virtually identical results.

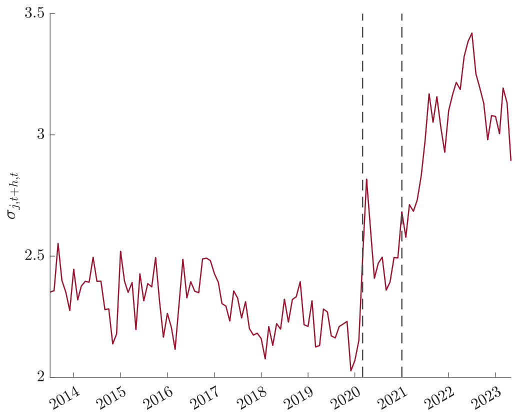
Figure A.18: Belief updating across groups: $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$: alternative calibration for prior expectations



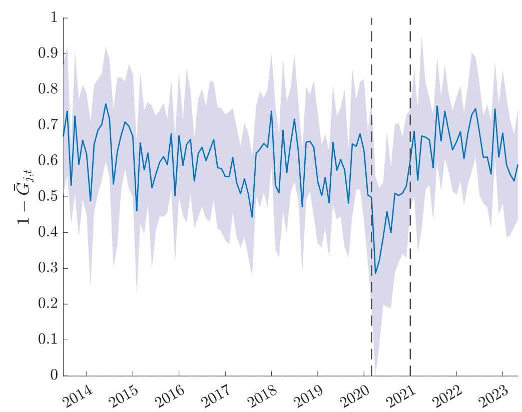
Notes: The figure reports the group-specific belief stickiness from (9), $\beta_1 + \mathbf{X}_{i,t}\mathbf{B}_3$, estimated including socioeconomic groups individually in the regression. Sample: 2013M6-2023M5. Bars represent the 95% confidence interval.

Figure A.19: Belief stickiness pre- and post-pandemic: alternative calibration for prior expectations

(a) Average uncertainty

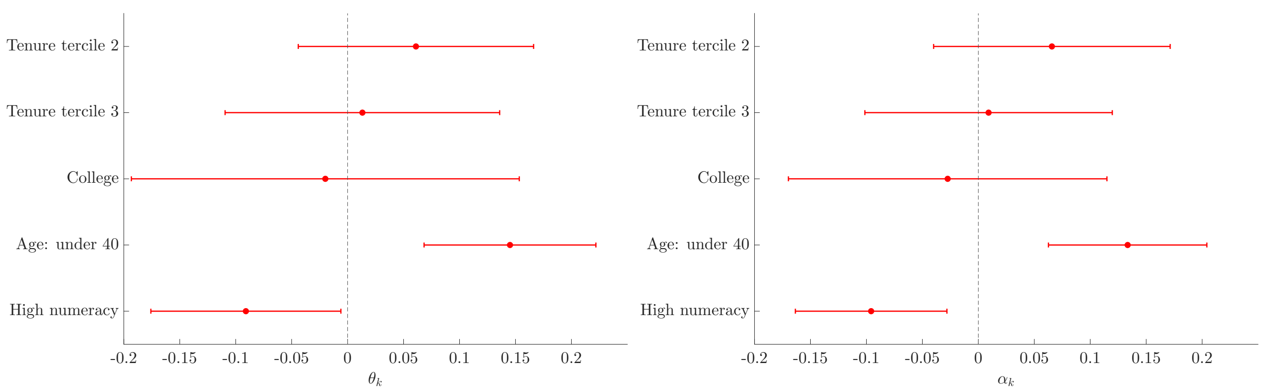


(b) Average belief stickiness



Notes: The left panel plots the average posterior belief uncertainty (standard deviation) across all consumers. The right panel plots the average belief stickiness across groups from regression 7, while the dashed blue lines represent the average 90% confidence interval. Standard errors are computed with a two-level bootstrap, within and across *(j,t)* groups. The first red vertical line corresponds to the start of Covid-19 in March 2020. The second red vertical line corresponds to the start of the high-inflation period in January 2021. Sample period: 2013M1 - 2023M5.

Figure A.20: Heterogeneity in overreaction bias: alternative calibration for prior expectations



(a) Heterogeneity in θ

(b) Heterogeneity in α

Notes: The left panel plots the estimated interactions with socioeconomic dummies and θ in 15. The right panel plots the estimated interactions with socioeconomic dummies and α in 15. Confidence intervals are at the 90% level and standard errors are computed with a two-level bootstrap, within and across *(j,t)* groups. Sample Jun13-Feb20.